



PT. SALIPI NAGARI PERKASA

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LAPORAN UJI MINYAK TRAFO

di

UNIT JASA PEMBANGKIT

JERANJANG



Yang dilakukan pada tanggal 25 September 2018

Untuk keadaan trafo bulan SEPTEMBER 2018

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1.0 Specific Information:

No.	Unit No.	Serial Number	TC	Test Requested
1.	MAIN TRAFO UNIT 1	100691	4	Acidity, Dielectric, Color, Visualization and Water Content
2.	TRAFO 1 ESP UNIT 1	10937	5	Acidity, Dielectric, Color, Visualization and Water Content
3.	TRAFO 2 ESP UNIT 1	10035	6	Acidity, Dielectric, Color, Visualization and Water Content
4.	TRAFO 3 ESP UNIT 1	10031	7	Acidity, Dielectric, Color, Visualization and Water Content
5.	TRAFO 4 ESP UNIT 1	10034	8	Acidity, Dielectric, Color, Visualization and Water Content
6.	MAIN TRAFO UNIT 3	1122170	9	Acidity, Dielectric, Color, Visualization and Water Content
7.	UAT 3	1035082	10	Acidity, Dielectric, Color, Visualization and Water Content
8.	TRAFO 1 ESP UNIT 3	0073084208	11	Acidity, Dielectric, Color, Visualization and Water Content
9.	TRAFO 2 ESP UNIT 3	0070084208	12	Acidity, Dielectric, Color, Visualization and Water Content
10.	TRAFO 3 ESP UNIT 3	0072084208	13	Acidity, Dielectric, Color, Visualization and Water Content
11.	TRAFO 4 UNIT 3	0071084208	14	Acidity, Dielectric, Color, Visualization and Water Content
12.	ASH HANDLING-1	10820108	15	Acidity, Dielectric, Color, Visualization and Water Content
13.	ASH HANDLING-2	10820107	16	Acidity, Dielectric, Color, Visualization and Water Content
14.	TRAFO 1 ESP UNIT 2	10037	17	Acidity, Dielectric, Color, Visualization and Water Content
15.	TRAFO 2 ESP UNIT 2	10036	18	Acidity, Dielectric, Color, Visualization and Water Content
16.	TRAFO 3 ESP UNIT 2	10033	19	Acidity, Dielectric, Color, Visualization and Water Content
17.	TRAFO 4 ESP UNIT 2	10032	20	Acidity, Dielectric, Color, Visualization and Water Content

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No.	Unit No.	Serial Number	TC	Test Requested
18.	MAIN TRAFO UNIT 2	1006902	21	Acidity, Dielectric, Color, Visualization and Water Content

Note:

* OLTT+ consists of Liquid Screen Tests, Dissolved Gas Analysis (DGA), Water Content, Furan Analysis, Metals-in-Oil Analysis, Liquid Power Factors, and Inhibitor Content (DBPC).

** Liquid Screen Test consists of Acidity, Interfacial Tension (IFT), Dielectric, Colour, Specific Gravity, Visualization, and Sediment.

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2.0 Our Comments and Recommendations of The Test Results Analysis:

We had reviewed each laboratory test result against respective benchmark parameter and following are our comments and recommendations.

No.	Unit No.	Serial Number	TC
1.	MAIN TRAFO UNIT 1	100691	4

A) Analysis

- **Acidity** of 0.014mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 96kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 7ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.46% M/dw is graded as **A**. For this transformer, it is recommended to maintain water content in oil at less than 20ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

4

DATE PRINTED 10/10/2018

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **MAIN TRAFU UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
30/08/2017	OFFLINE	49	49	NORMAL	FAIR	GOOD	NONE
29/09/2017	OFFLINE	48		NORMAL	FAIR	GOOD	NONE
30/10/2017	ONLINE	60	64	NORMAL	FAIR	GOOD	NONE
27/11/2017	ONLINE		51				
06/12/2017	OFFLINE	50	48				
22/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE	50	55				
24/07/2018	ONLINE					GOOD	
25/08/2018	OFFLINE						
25/09/2018	ONLINE						

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
30/08/2017	ROUTINE	0.017 AC		92 AC	0.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.014 AC		82 AC	0.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.014 AC		79 AC	0.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.014 AC		53 AC	0.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.011 AC	40.2 AC	87 AC	0.5 AC	0.8745 AC	CLEAR AC	
22/05/2018	ROUTINE	0.011 AC		77 AC	0.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.014 AC	41.9 AC	69 AC	0.5 AC	0.8716 AC	CLEAR AC	
24/07/2018	ROUTINE	0.014 AC		86 AC	0.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.017 AC		90 AC	0.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.014 AC		96 AC	0.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	147	9471	43278	32	655	2728	4	5	ND	843	56320
06/12/2017	183	12895	43794	39	715	3166	5	5	ND	947	60802
28/06/2018	176	11044	46776	41	758	3393	5	6	ND	986	62199

RECOMMENDATION:

NO FURTHER GENERATION OF HYDROGEN IS NOTED. HOWEVER, INCREASING PATTERN OF CARBON OXIDES ARE DETECTED.

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 100691
KVA	- 40000
PRI	- 165000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.1%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30857
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
30/08/2017	34	6	0.49	A
29/09/2017	36	8	0.57	A
30/10/2017	44	10	0.45	A
27/11/2017	42	8	0.40	A
06/12/2017	38	7	0.46	A
22/05/2018	30	10	1.05	B
28/06/2018	41	7	0.35	A
24/07/2018	32	6	0.57	A
25/08/2018	30	6	0.63	A
25/09/2018	38	7	0.46	A

FIELD SERVICE

DATE	SERVICE
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CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

4

DATE PRINTED 10/10/2018

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **MAIN TRAFU UNIT 1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 100691
KVA	- 40000
PRI	- 165000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.1%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30857
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.02	0.01	0.01
06/12/2017	ND	0.01	0.04
28/06/2018	ND	ND	0.04

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.140 AC
06/12/2017	0.000 AC	0.040 AC
28/06/2018	0.000 AC	0.140 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.24 AC
06/12/2017	0.27 AC
28/06/2018	0.24 AC

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA
06/12/2017	NON-COR	2C	LAVENDER BLUE
28/06/2018	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
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No.	Unit No.	Serial Number	TC
2.	TRAFO 1 ESP UNIT 1	10937	5

A) Analysis

- **Acidity** of 0.017mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 63kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water content** in oil of 18ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.28% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

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DATE PRINTED 10/10/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
30/08/2017	OFFLINE	30				GOOD	NONE
29/09/2017	OFFLINE	35		NORMAL		GOOD	NONE
30/10/2017	OFFLINE	50		NORMAL		GOOD	NONE
27/11/2017	ONLINE						
06/12/2017	OFFLINE						
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE						
24/07/2018	OFFLINE						
25/08/2018	OFFLINE						
25/09/2018	ONLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10937
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
30/08/2017	ROUTINE	0.056 QU		65 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.059 QU		58 AC	1.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.059 QU		23 UN	1.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.062 QU		25 QU	1.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.062 QU	25.0 UN	48 AC	1.5 AC	0.8583 AC	CLEAR AC	
21/05/2018	ROUTINE	0.065 QU		38 AC	1.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.067 QU	25.9 UN	32 AC	1.5 AC	0.8560 AC	CLEAR AC	
24/07/2018	ROUTINE	0.073 QU		38 AC	1.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.020 AC		71 AC	0.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.017 AC		63 AC	0.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	35	20494	67487	4	69	1143	ND	1	ND	109	89233
06/12/2017	39	19933	67198	4	83	1261	ND	1	ND	127	88519
28/06/2018	44	20039	64529	4	119	1696	ND	2	ND	169	86433

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
30/08/2017	29	18	1.89	D
29/09/2017	32	19	1.80	D
30/10/2017	42	32	1.60	C
27/11/2017	41	33	1.65	C
06/12/2017	30	18	1.89	D
21/05/2018	31	33	3.13	E
28/06/2018	32	30	2.85	E
24/07/2018	29	19	2.00	D
25/08/2018	32	12	1.14	B
25/09/2018	35	18	1.28	B

FIELD SERVICE

DATE SERVICE
01/08/2018 CHANGED OIL

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

5

DATE PRINTED 10/10/2018

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10937
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	95	ND	ND	ND	95
06/12/2017	ND	75	ND	ND	ND	75
28/06/2018	ND	85	ND	ND	ND	85

RECOMMENDATION:
FURAN ANALYSIS SHOWS DETECTION OF 2FOL WHICH INDICATES THAT THE SOLID INSULATION IS WET. IT IS ADVISABLE TO MAINTAIN THE OIL IN A GOOD CONDITION SO AS NOT TO FURTHER DETERIORATE THE PAPER INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	0.01
06/12/2017	ND	ND	ND
28/06/2018	ND	ND	0.03

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.870 AC
06/12/2017	0.000 AC	0.570 AC
28/06/2018	0.000 AC	0.680 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	ND UN
06/12/2017	ND UN
28/06/2018	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1B	DARK ORANGE
06/12/2017	NON-COR	1A	LIGHT ORANGE
28/06/2018	NON-COR	1B	DARK ORANGE

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE SERVICE
01/08/2018 CHANGED OIL

No.	Unit No.	Serial Number	TC
3.	TRAFO 2 ESP UNIT 1	10035	6

A) Analysis

- **Acidity** of 0.020mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 67kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water content** in oil of 11ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.04% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

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DATE PRINTED 10/10/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
30/08/2017	OFFLINE	39		NORMAL	FAIR	GOOD	NONE
29/09/2017	OFFLINE	40		NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE	44		NORMAL	FAIR	GOOD	NONE
27/11/2017	ONLINE						
06/12/2017	OFFLINE						
21/05/2018	OFFLINE			LOW		GOOD	NONE
28/06/2018	ONLINE						
24/07/2018	OFFLINE						
25/08/2018	OFFLINE						
25/09/2018	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10035
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
30/08/2017	ROUTINE	0.042 AC		42 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.048 AC		59 AC	1.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.051 QU		62 AC	1.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.051 QU		50 AC	1.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.053 QU	27.3 UN	51 AC	1.5 AC	0.8589 AC	CLEAR AC	
21/05/2018	ROUTINE	0.056 QU		45 AC	1.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.059 QU	27.6 UN	55 AC	1.5 AC	0.8570 AC	CLEAR AC	
24/07/2018	ROUTINE	0.062 QU		47 AC	1.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.022 AC		56 AC	0.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.020 AC		67 AC	0.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	27	18639	66781	3	64	1106	ND	1	ND	95	86621
06/12/2017	20	19360	61590	3	45	728	ND	1	ND	69	81747
28/06/2018	22	22494	64443	4	57	789	ND	ND	ND	83	87809

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
30/08/2017	28	14	1.65	C
29/09/2017	31	19	1.80	D
30/10/2017	35	17	1.21	B
27/11/2017	35	18	1.28	B
06/12/2017	30	17	1.78	D
21/05/2018	31	20	1.90	D
28/06/2018	32	17	1.62	C
24/07/2018	30	15	1.58	C
25/08/2018	31	13	1.24	B
25/09/2018	31	11	1.04	B

FIELD SERVICE

DATE SERVICE
01/08/2018 CHANGED OIL

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

6

DATE PRINTED 10/10/2018

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10035
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	62	ND	ND	ND	62
06/12/2017	ND	67	ND	ND	ND	67
28/06/2018	ND	84	ND	ND	ND	84

RECOMMENDATION:
FURAN ANALYSIS SHOWS DETECTION OF 2FOL WHICH INDICATES THAT THE SOLID INSULATION IS WET. IT IS ADVISABLE TO MAINTAIN THE OIL IN A GOOD CONDITION SO AS NOT TO FURTHER DETERIORATE THE PAPER INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	ND
06/12/2017	ND	ND	0.02
28/06/2018	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.900 AC
06/12/2017	0.000 AC	0.980 AC
28/06/2018	0.000 AC	1.310 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.02 UN
06/12/2017	0.02 UN
28/06/2018	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1A	LIGHT ORANGE
06/12/2017	NON-COR	1B	DARK ORANGE
28/06/2018	NON-COR	1B	DARK ORANGE

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE SERVICE
01/08/2018 CHANGED OIL

No.	Unit No.	Serial Number	TC
4.	TRAFO 3 ESP UNIT 1	10031	7

A) Analysis

- **Acidity** of 0.051mgKOH/gm is **QUESTIONABLE**. The levels remained stable without further oil degradation. Stabilized condition may proceed with annual monitoring.
- **Dielectric** of 46kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water content** in oil of 21ppm is **QUESTIONABLE**. Meanwhile Water Content in solid insulation of 2.00% M/dw is graded as **D**. For this transformer, it is advisable to maintain the water content level below 20ppm. This indicates the presence of high water content in the oil.

B) Recommendation

- Based on previous oil analysis, the IFT of the oil is unfavorable. Together with the high level of Acidity and Water Content, it is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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DATE PRINTED 10/10/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
30/08/2017	OFFLINE	35		NORMAL	FAIR	GOOD	NONE
29/09/2017	OFFLINE	38		NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE	44		NORMAL	FAIR	GOOD	NONE
27/11/2017	ONLINE						
06/12/2017	OFFLINE						
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE						
24/07/2018	OFFLINE						
25/08/2018	OFFLINE						
25/09/2018	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10031
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
30/08/2017	ROUTINE	0.042 AC		61 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.045 AC		72 AC	1.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.048 AC		72 AC	1.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.051 QU		62 AC	1.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.053 QU	28.3 QU	57 AC	1.5 AC	0.8624 AC	CLEAR AC	
21/05/2018	ROUTINE	0.056 QU		48 AC	1.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.053 QU	29.7 QU	62 AC	1.5 AC	0.8605 AC	CLEAR AC	
24/07/2018	ROUTINE	0.051 QU		64 AC	1.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.056 QU		57 AC	1.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.051 QU		46 AC	1.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	32	15273	65111	5	95	1002	ND	2	ND	134	81520
06/12/2017	7	14730	65738	3	42	617	ND	ND	ND	52	81137
28/06/2018	9	17958	66007	4	46	781	ND	1	ND	60	84806

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
30/08/2017	30	13	1.37	C
29/09/2017	32	14	1.33	B
30/10/2017	37	16	1.04	B
27/11/2017	34	17	1.39	C
06/12/2017	30	16	1.68	C
21/05/2018	31	21	2.00	D
28/06/2018	30	17	1.78	D
24/07/2018	30	17	1.78	D
25/08/2018	30	15	1.58	C
25/09/2018	32	21	2.00	D

FIELD SERVICE

DATE SERVICE

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

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DATE PRINTED 10/10/2018

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10031
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	32	ND	ND	ND	32
06/12/2017	ND	14	ND	ND	ND	14
28/06/2018	ND	20	ND	ND	ND	20

RECOMMENDATION:
FURAN ANALYSIS SHOWS MINOR DETECTION OF FURANIC COMPOUNDS. THIS IS IN LINE WITH A
HEALTHY SOLID INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.01	0.01	0.01
06/12/2017	ND	ND	ND
28/06/2018	ND	ND	0.11

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.700 AC
06/12/2017	0.000 AC	0.630 AC
28/06/2018	0.000 AC	1.030 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.01 UN
06/12/2017	0.01 UN
28/06/2018	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1A	LIGHT ORANGE
06/12/2017	NON-COR	2C	LAVENDER BLUE
28/06/2018	NON-COR	1B	DARK ORANGE

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
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No.	Unit No.	Serial Number	TC
5.	TRAFO 4 ESP UNIT 1	10034	8

A) Analysis

- **Acidity** of 0.014mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 45kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water content** in oil of 20ppm is **QUESTIONABLE**. Meanwhile Water Content in solid insulation of 2.10% M/dw is graded as **D**. For this transformer, it is advisable to maintain the water content level below 20ppm. This indicates the presence of high water content in the oil.

B) Recommendation

- Based on previous oil analysis, the IFT of the oil is unfavorable. Together with the high level Water Content, it is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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DATE PRINTED 10/10/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
30/08/2017	OFFLINE	34		NORMAL	FAIR	GOOD	NONE
29/09/2017	OFFLINE	40		NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE	44		NORMAL	FAIR	GOOD	NONE
27/11/2017	ONLINE						
06/12/2017	OFFLINE						
21/05/2018	OFFLINE			LOW	FAIR	GOOD	NONE
28/06/2018	ONLINE						
24/07/2018	OFFLINE						
25/08/2018	OFFLINE						
25/09/2018	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10034
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
30/08/2017	ROUTINE	0.028 AC		44 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.031 AC		53 AC	1.0 AC		CLEAR AC	
30/10/2017	ROUTINE	0.028 AC		48 AC	1.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.034 AC		41 AC	1.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.031 AC	27.2 UN	41 AC	1.5 AC	0.8645 AC	CLEAR AC	
21/05/2018	ROUTINE	0.034 AC		37 AC	1.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.037 AC	27.5 UN	55 AC	1.5 AC	0.8635 AC	CLEAR AC	
24/07/2018	ROUTINE	0.039 AC		50 AC	1.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.017 AC		61 AC	0.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.014 AC		45 AC	0.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	13	12265	66770	5	172	1206	ND	8	ND	198	80439
06/12/2017	8	13331	63527	3	70	720	ND	1	ND	82	77660
28/06/2018	9	17962	64248	4	75	1054	ND	2	ND	90	83354

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
30/08/2017	30	13	1.37	C
29/09/2017	33	15	1.23	B
30/10/2017	39	18	0.99	B
27/11/2017	37	30	1.95	D
06/12/2017	30	18	1.89	D
21/05/2018	31	23	2.18	D
28/06/2018	30	20	2.10	D
24/07/2018	31	19	1.80	D
25/08/2018	30	14	1.47	C
25/09/2018	30	20	2.10	D

FIELD SERVICE

DATE SERVICE
01/08/2018 CHANGED OIL

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

8

DATE PRINTED 10/10/2018

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10034
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	18	ND	ND	ND	18
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	21	ND	ND	ND	21

RECOMMENDATION:
FURAN ANALYSIS SHOWS MINOR DETECTION OF FURANIC COMPOUNDS. THIS IS IN LINE WITH A
HEALTHY SOLID INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.05	0.02	0.01
06/12/2017	ND	ND	0.02
28/06/2018	ND	ND	0.05

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.180 QU	6.680 UN
06/12/2017	0.010 AC	2.180 AC
28/06/2018	0.060 AC	2.850 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.08 UN
06/12/2017	0.06 UN
28/06/2018	0.02 UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	2C	LAVENDER BLUE
06/12/2017	NON-COR	2C	LAVENDER BLUE
28/06/2018	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE SERVICE
01/08/2018 CHANGED OIL

No.	Unit No.	Serial Number	TC
6.	MAIN TRAF0 UNIT 3	1122170	9

A) Analysis

- **Acidity** of 0.017mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 83kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 6ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.43% M/dw is graded as **A**. For this transformer, it is recommended to maintain water content in oil at less than 20ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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DATE PRINTED 10/10/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAFU UNIT 3

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
30/08/2017	ONLINE	55	60	NORMAL	FAIR	GOOD	NONE
29/09/2017	ONLINE	58	68	NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE	32	32	NORMAL	FAIR	GOOD	NONE
27/11/2017	ONLINE	46	50				
06/12/2017	ONLINE	57	48				
22/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE	48	50	NORMAL			
24/07/2018	ONLINE						
25/08/2018	ONLINE						
25/09/2018	ONLINE						

NAMEPLATE DATA

MFG	- SHANDONG
DATE MFG	- 2010
MAIN TANK S/N	- 1122170
KVA	- 40000
PRI	- 165000 V
SEC	- 6300 V
KILOGRAMS	- 73920
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAF
IMPEDENCE	- 13.79%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 26708

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
30/08/2017	ROUTINE	0.017 AC		86 AC	0.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.014 AC		86 AC	0.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.014 AC		88 AC	0.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.014 AC		87 AC	0.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.014 AC	41.5 AC	84 AC	0.5 AC	0.8832 AC	CLEAR AC	
22/05/2018	ROUTINE	0.014 AC		58 AC	0.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.014 AC	41.0 AC	85 AC	0.5 AC	0.8805 AC	CLEAR AC	
24/07/2018	ROUTINE	0.014 AC		77 AC	0.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.017 AC		82 AC	0.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.017 AC		83 AC	0.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	24	10131	44084	11	641	2250	1	1	ND	678	57143
06/12/2017	39	14023	43226	11	668	2456	ND	1	ND	719	60424
28/06/2018	18	11431	48283	13	700	2512	1	1	ND	733	62959

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
CARBON MONOXIDE AND CARBON DIOXIDE LEVEL ARE HIGH. OTHER GASES LEVELS ARE ASSESSED AS ACCEPTABLE.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
30/08/2017	35	8	0.57	A
29/09/2017	42	10	0.50	A
30/10/2017	28	9	1.06	B
27/11/2017	35	8	0.57	A
06/12/2017	37	9	0.58	A
22/05/2018	31	11	1.04	B
28/06/2018	36	8	0.57	A
24/07/2018	33	8	0.66	A
25/08/2018	32	7	0.67	A
25/09/2018	36	6	0.43	A

FIELD SERVICE

DATE	SERVICE
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DATE PRINTED 10/10/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAFU UNIT 3

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- SHANDONG
DATE MFG	- 2010
MAIN TANK S/N	- 1122170
KVA	- 40000
PRI	- 165000 V
SEC	- 6300 V
KILOGRAMS	- 73920
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAF
IMPEDENCE	- 13.79%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 26708
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
04/07/2017	0.01	0.01	0.02
06/12/2017	0.01	ND	0.03
28/06/2018	ND	0.01	0.06

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.130 AC
06/12/2017	0.000 AC	0.070 AC
28/06/2018	0.000 AC	0.110 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.22 AC
06/12/2017	0.22 AC
28/06/2018	0.19 QU

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA
06/12/2017	NON-COR	2C	LAVENDER BLUE
28/06/2018	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
7.	UAT 3	1035082	10

A) Analysis

- **Acidity** of 0.020mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 75kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 8ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.57% M/dw is graded as **A**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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DATE PRINTED 10/10/2018
ID# - 601669

CUST - PT IP UJP PLTU JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **UAT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
30/08/2017	ONLINE		58	NORMAL	FAIR	GOOD	NONE
29/09/2017	ONLINE		58	NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE		44	NORMAL	FAIR	GOOD	NONE
27/11/2017	ONLINE		60				
06/12/2017	ONLINE		60				
22/05/2018	ONLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE		50				
24/07/2018	ONLINE						
25/08/2018	ONLINE						
25/09/2018	ONLINE						

NAMEPLATE DATA

MFG	- SHANDONG
DATE MFG	- 2010
MAIN TANK S/N	- 1035082
KVA	- 6300
PRI	- 6300 V
SEC	- 6300 V
KILOGRAMS	- 14600
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.78%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 3749

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
30/08/2017	ROUTINE	0.022 AC		89 AC	0.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.020 AC		80 AC	0.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.022 AC		86 AC	0.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.022 AC		86 AC	0.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.020 AC	37.6 AC	86 AC	0.5 AC	0.8820 AC	CLEAR AC	
22/05/2018	ROUTINE	0.020 AC		81 AC	0.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.022 AC	37.5 AC	79 AC	0.5 AC	0.8793 AC	CLEAR AC	
24/07/2018	ROUTINE	0.020 AC		76 AC	0.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.022 AC		78 AC	0.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.020 AC		75 AC	0.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	20	16028	68308	20	971	3536	5	7	ND	1023	88895
06/12/2017	36	18752	66758	19	1116	3622	5	7	ND	1183	90315
28/06/2018	35	13798	68160	22	1144	3644	5	7	1	1214	86816

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:

RETEST 6 MONTHS: THE DETECTION OF ACETYLENE IS SOME CAUSE FOR CONCERN. CLOSE MONITORING IS NEEDED TO DIAGNOSE THE FAULT

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
30/08/2017	35	11	0.78	A
29/09/2017	37	12	0.78	A
30/10/2017	29	11	1.15	A
27/11/2017	37	12	0.78	A
06/12/2017	39	12	0.66	A
22/05/2018	29	14	1.47	B
28/06/2018	33	10	0.82	A
24/07/2018	32	11	1.04	A
25/08/2018	29	8	0.84	A
25/09/2018	35	8	0.57	A

FIELD SERVICE

DATE	SERVICE
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DATE PRINTED 10/10/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - UAT 3

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- SHANDONG
DATE MFG	- 2010
MAIN TANK S/N	- 1035082
KVA	- 6300
PRI	- 6300 V
SEC	- 6300 V
KILOGRAMS	- 14600
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.78%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 3749
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.02	0.01	0.04
06/12/2017	ND	0.01	0.08
28/06/2018	ND	0.01	0.11

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.390 AC
06/12/2017	0.000 AC	0.170 AC
28/06/2018	0.000 AC	0.140 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.23 AC
06/12/2017	0.23 AC
28/06/2018	0.21 AC

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA
06/12/2017	NON-COR	2C	LAVENDER BLUE
28/06/2018	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
8.	TRAFO 1 ESP UNIT 3	0073084208	11

A) Analysis

- **Acidity** of 0.020mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 64kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water content** in oil of 10ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.05% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

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DATE PRINTED 10/10/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
30/08/2017	ONLINE			NORMAL	FAIR	GOOD	NONE
29/09/2017	ONLINE			NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE			NORMAL	FAIR	GOOD	NONE
28/11/2017	ONLINE						
06/12/2017	ONLINE						
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	OFFLINE						
24/07/2018	OFFLINE						
25/08/2018	OFFLINE						
25/09/2018	OFFLINE						

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0073084208
KVA	-
PRI	- 72000 V
SEC	- 416 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
30/08/2017	ROUTINE	0.028 AC		36 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.025 AC		45 AC	1.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.025 AC		40 AC	1.5 AC		CLEAR AC	
28/11/2017	ROUTINE	0.025 AC		31 AC	1.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.022 AC	34.6 AC	31 AC	1.5 AC	0.8180 QU	CLEAR AC	
21/05/2018	ROUTINE	0.022 AC		31 AC	0.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.022 AC	35.0 AC	60 AC	0.5 AC	0.8154 QU	CLEAR AC	
24/07/2018	ROUTINE	0.020 AC		53 AC	0.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.022 AC		77 AC	0.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.020 AC		64 AC	0.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	14	12943	68807	3	62	1243	ND	ND	ND	79	83072
06/12/2017	14	12739	65097	3	66	612	ND	ND	ND	83	78531
28/06/2018	19	15552	64448	3	57	1391	ND	ND	ND	79	81470

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
30/08/2017	30	22	2.31	D
29/09/2017	34	20	1.64	C
30/10/2017	28	19	2.24	D
28/11/2017	43	34	1.53	C
06/12/2017	43	31	1.39	C
21/05/2018	31	27	2.57	D
28/06/2018	32	10	0.95	B
24/07/2018	26	9	1.17	B
25/08/2018	26	9	1.17	B
25/09/2018	30	10	1.05	B

FIELD SERVICE

DATE	SERVICE
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CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

11

DATE PRINTED 10/10/2018

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0073084208
KVA	-
PRI	- 72000 V
SEC	- 416 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.01	0.01	ND
06/12/2017	0.01	ND	ND
28/06/2018	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.350 AC
06/12/2017	0.000 AC	0.040 AC
28/06/2018	0.000 AC	0.050 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	ND UN
06/12/2017	0.18 QU
28/06/2018	0.16 QU

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1B	DARK ORANGE
06/12/2017	NON-COR	2E	BRASSY GOLD
28/06/2018	NON-COR	1B	DARK ORANGE

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
9.	TRAFO 2 ESP UNIT 3	0070084208	12

A) Analysis

- **Acidity** of 0.22mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 31kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 32ppm is **UNACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.76% M/dw is graded as **D**. For this transformer, it is recommended to maintain water content in oil at less than 20ppm.

B) Recommendation

- Based on previous oil analysis, the IFT of the oil is unfavorable. Together with the high level of Water Content, it is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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DATE PRINTED 10/10/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
30/08/2017	OFFLINE			NORMAL	FAIR	GOOD	NONE
29/09/2017	ONLINE			NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE			NORMAL	FAIR	GOOD	NONE
28/11/2017	ONLINE						
06/12/2017	ONLINE						
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	OFFLINE						
24/07/2018	ONLINE						
25/08/2018	OFFLINE						
25/09/2018	ONLINE						

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0070084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
30/08/2017	ROUTINE	0.109 UN		34 AC	2.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.115 UN		21 UN	2.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.118 UN		33 AC	2.5 AC		CLEAR AC	
28/11/2017	ROUTINE	0.017 AC		25 QU	0.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.022 AC	29.7 QU	29 QU	0.5 AC	0.8106 QU	CLEAR AC	
21/05/2018	ROUTINE	0.022 AC		28 QU	0.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.025 AC	30.2 QU	34 AC	0.5 AC	0.8080 QU	CLEAR AC	
24/07/2018	ROUTINE	0.022 AC		30 AC	0.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.025 AC		67 AC	0.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.022 AC		31 AC	0.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	38	16781	70523	5	129	2431	3	5	ND	180	89915
06/12/2017	42	19523	66059	3	102	912	ND	ND	ND	147	86641
28/06/2018	43	22272	68719	4	38	1226	ND	ND	ND	85	92302

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
30/08/2017	26	17	2.21	D
29/09/2017	36	37	2.63	D
30/10/2017	28	20	2.36	D
28/11/2017	43	28	1.26	B
06/12/2017	43	44	1.98	D
21/05/2018	31	30	2.85	E
28/06/2018	30	16	1.68	C
24/07/2018	32	25	2.38	D
25/08/2018	26	11	1.43	C
25/09/2018	39	32	1.76	D

FIELD SERVICE

DATE SERVICE
01/07/2018 PURIFIED

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

12

DATE PRINTED 10/10/2018

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	169	ND	ND	169
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0070084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	0.01
06/12/2017	ND	ND	ND
28/06/2018	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.330 AC
06/12/2017	0.000 AC	0.030 AC
28/06/2018	0.000 AC	0.040 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	ND UN
06/12/2017	0.20 AC
28/06/2018	0.19 QU

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	2C	LAVENDER BLUE
06/12/2017	NON-COR	3A	MAGENTA
28/06/2018	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE SERVICE
01/07/2018 PURIFIED

No.	Unit No.	Serial Number	TC
10.	TRAFO 3 ESP UNIT 3	0072084208	13

A) Analysis

- **Acidity** of 0.020mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 47kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water content** in oil of 20ppm is **QUESTIONABLE**. Meanwhile Water Content in solid insulation of 1.10% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm. This indicates the presence of high water content in the oil.

B) Recommendation

- Based on previous oil analysis, the Specific Gravity of the oil needs further investigation. Should the unit is safe to operate with the current oil-filled, highly recommended to perform **OIL PURIFICATION**. Result indicates that the moisture content in the transformer oil is not conducive for normal operation of a transformer. Oil purification will remove excessive moisture detected in the transformer oil and thus improving dielectric breakdown voltage of the mineral oil. Furthermore, passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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DATE PRINTED 10/10/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
30/08/2017	ONLINE			NORMAL	FAIR	GOOD	NONE
29/09/2017	ONLINE			NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE			LOW	FAIR	GOOD	NONE
28/11/2017	ONLINE						
06/12/2017	ONLINE						
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE						
24/07/2018	ONLINE						
25/08/2018	OFFLINE						
25/09/2018	ONLINE						

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0072084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
30/08/2017	ROUTINE	0.025 AC		50 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.028 AC		59 AC	1.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.028 AC		55 AC	1.5 AC		CLEAR AC	
28/11/2017	ROUTINE	0.014 AC		39 AC	0.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.014 AC	39.8 AC	45 AC	0.5 AC	0.8099 QU	CLEAR AC	
21/05/2018	ROUTINE	0.014 AC		54 AC	0.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.017 AC	38.4 AC	43 AC	0.5 AC	0.8075 QU	CLEAR AC	
24/07/2018	ROUTINE	0.014 AC		43 AC	0.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.014 AC		76 AC	0.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.020 AC		47 AC	0.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	21	16364	70694	3	38	1068	ND	ND	ND	62	88188
06/12/2017	15	14515	65277	2	40	559	ND	ND	ND	57	80408
28/06/2018	30	15202	64188	4	65	1833	ND	ND	ND	99	81322

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
30/08/2017	35	21	1.49	C
29/09/2017	37	22	1.43	C
30/10/2017	28	16	1.89	D
28/11/2017	44	24	1.08	B
06/12/2017	44	29	1.30	B
21/05/2018	32	23	2.18	D
28/06/2018	35	22	1.56	C
24/07/2018	34	20	1.64	C
25/08/2018	25	11	1.43	C
25/09/2018	40	20	1.10	B

FIELD SERVICE

DATE SERVICE
01/07/2018 PURIFIED

DATE PRINTED 10/10/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - TRAFO 3 ESP UNIT 3

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0072084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	4	ND	ND	ND	4

RECOMMENDATION:
FURAN ANALYSIS SHOWS MINOR DETECTION OF FURANIC COMPOUNDS. THIS IS IN LINE WITH A
HEALTHY SOLID INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	ND
06/12/2017	ND	ND	ND
28/06/2018	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.620 AC
06/12/2017	0.000 AC	0.080 AC
28/06/2018	0.000 AC	0.010 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.01 UN
06/12/2017	0.21 AC
28/06/2018	0.20 AC

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	2C	LAVENDER BLUE
06/12/2017	NON-COR	2E	BRASSY GOLD
28/06/2018	NON-COR	2E	BRASSY GOLD

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE SERVICE
01/07/2018 PURIFIED

No.	Unit No.	Serial Number	TC
11.	TRAFO 4 UNIT 3	0071084208	14

A) Analysis

- **Acidity** of 0.022mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 51kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water content** in oil of 11ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.15% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.

B) Recommendation

- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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DATE PRINTED 10/10/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
30/08/2017	OFFLINE			NORMAL	FAIR	GOOD	NONE
29/09/2017	OFFLINE			NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE			NORMAL	FAIR	GOOD	NONE
28/11/2017	OFFLINE						
06/12/2017	OFFLINE						
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE						
24/07/2018	OFFLINE						
25/08/2018	OFFLINE						
25/09/2018	ONLINE						

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0071084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
30/08/2017	ROUTINE	0.020 AC		55 AC	1.0 AC		CLEAR AC	
29/09/2017	ROUTINE	0.022 AC		59 AC	1.0 AC		CLEAR AC	
30/10/2017	ROUTINE	0.025 AC		55 AC	1.0 AC		CLEAR AC	
28/11/2017	ROUTINE	0.022 AC		69 AC	1.0 AC		CLEAR AC	
06/12/2017	ROUTINE	0.025 AC	29.8 QU	43 AC	1.0 AC	0.8471 AC	CLEAR AC	
21/05/2018	ROUTINE	0.028 AC		37 AC	1.0 AC		CLEAR AC	
28/06/2018	ROUTINE	0.025 AC	30.5 QU	71 AC	1.0 AC	0.8442 AC	CLEAR AC	
24/07/2018	ROUTINE	0.022 AC		42 AC	1.0 AC		CLEAR AC	
25/08/2018	ROUTINE	0.025 AC		66 AC	1.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.022 AC		51 AC	1.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	10	18669	66036	3	24	665	ND	ND	ND	37	85407
06/12/2017	14	15414	63712	2	22	675	ND	ND	ND	38	79839
28/06/2018	20	14690	67822	2	22	683	ND	ND	ND	44	83239

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
30/08/2017	27	17	2.01	D
29/09/2017	29	15	1.58	C
30/10/2017	28	15	1.77	D
28/11/2017	29	16	1.68	C
06/12/2017	32	19	1.80	D
21/05/2018	30	21	2.20	D
28/06/2018	36	11	0.78	A
24/07/2018	26	11	1.43	C
25/08/2018	27	11	1.30	B
25/09/2018	30	11	1.15	B

FIELD SERVICE

DATE SERVICE

DATE PRINTED 10/10/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 UNIT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0071084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	ND	ND
06/12/2017	ND	0.03	ND
28/06/2018	ND	0.03	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.360 AC
06/12/2017	0.000 AC	0.370 AC
28/06/2018	0.000 AC	0.550 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.01 UN
06/12/2017	0.03 UN
28/06/2018	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA
06/12/2017	NON-COR	1B	DARK ORANGE
28/06/2018	NON-COR	1B	DARK ORANGE

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
12.	ASH HANDLING-1	10820108	15

A) Analysis

- **Acidity** of 0.025mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 43kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 17ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.21% M/dw is graded as **A**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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DATE PRINTED 10/10/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **ASH HANDLING-1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
31/07/2017	ONLINE						
30/08/2017	OFFLINE	30		NORMAL	FAIR		NONE
29/09/2017	OFFLINE	40		NORMAL	FAIR		NONE
30/10/2017	ONLINE	45		NORMAL	FAIR		NONE
27/11/2017	ONLINE	46					
06/12/2017	ONLINE	52					
22/05/2018	ONLINE	45		NORMAL		GOOD	NONE
28/06/2018	ONLINE						
24/07/2018	ONLINE						
25/08/2018	OFFLINE						

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820108
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.56%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
30/08/2017	ROUTINE	0.022 AC		53 AC	1.0 AC		CLEAR AC	
29/09/2017	ROUTINE	0.022 AC		57 AC	1.0 AC		CLEAR AC	
30/10/2017	ROUTINE	0.022 AC		45 AC	1.0 AC		CLEAR AC	
27/11/2017	ROUTINE	0.020 AC		36 AC	1.0 AC		CLEAR AC	
06/12/2017	ROUTINE	0.017 AC	37.5 AC	34 AC	1.0 AC	0.8832 AC	CLEAR AC	
22/05/2018	ROUTINE	0.022 AC		33 AC	1.0 AC		CLEAR AC	
28/06/2018	ROUTINE	0.025 AC	37.5 AC	42 AC	1.0 AC	0.8804 AC	CLEAR AC	
24/07/2018	ROUTINE	0.022 AC		39 AC	1.0 AC		CLEAR AC	
25/08/2018	ROUTINE	0.025 AC		60 AC	1.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.025 AC		43 AC	1.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	35	10984	71058	20	441	5037	7	8	ND	511	87590
06/12/2017	49	12034	67124	20	558	5440	8	8	ND	643	85241
28/06/2018	53	10939	69394	20	557	5425	9	8	ND	647	86405

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
GAS LEVELS HAVE REMAINED STABLE. PROCEED WITH ANNUAL MONITORING.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
30/08/2017	28	16	1.89	B
29/09/2017	34	16	1.31	B
30/10/2017	33	20	1.64	B
27/11/2017	36	29	2.06	C
06/12/2017	36	26	1.85	B
22/05/2018	33	26	2.13	C
28/06/2018	32	24	2.28	C
24/07/2018	32	20	1.90	B
25/08/2018	30	16	1.68	B
25/09/2018	35	17	1.21	A

FIELD SERVICE

DATE SERVICE

DATE PRINTED 10/10/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **ASH HANDLING-1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820108
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.56%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	0.01
06/12/2017	ND	ND	0.01
28/06/2018	ND	0.01	0.04

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.730 AC
06/12/2017	0.000 AC	0.240 AC
28/06/2018	0.000 AC	0.460 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.23 AC
06/12/2017	0.24 AC
28/06/2018	0.22 AC

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1B	DARK ORANGE
06/12/2017	NON-COR	2C	LAVENDER BLUE
28/06/2018	NON-COR	1B	DARK ORANGE

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
13.	ASH HANDLING-2	10820107	16

A) Analysis

- **Acidity** of 0.022mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 36kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 19ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 2.00% M/dw is graded as **B**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

Page Disclaimer :

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DATE PRINTED 10/10/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - ASH HANDLING-2

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
30/08/2017	ONLINE	47		NORMAL	FAIR		NONE
29/09/2017	ONLINE	60		NORMAL	FAIR		NONE
30/10/2017	OFFLINE	32		NORMAL	FAIR		NONE
27/11/2017	OFFLINE	40					
06/12/2017	ONLINE	48					
22/05/2018	ONLINE	45				GOOD	NONE
28/06/2018	OFFLINE						
24/07/2018	ONLINE						
25/08/2018	OFFLINE						
25/09/2018	OFFLINE						

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820107
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.6%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
30/08/2017	ROUTINE	0.022 AC		31 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.022 AC		28 QU	1.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.020 AC		46 AC	1.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.022 AC		47 AC	1.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.020 AC	36.6 AC	24 UN	1.5 AC	0.8840 AC	CLEAR AC	
22/05/2018	ROUTINE	0.020 AC		24 UN	1.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.022 AC	35.0 AC	50 AC	1.5 AC	0.8811 AC	CLEAR AC	
24/07/2018	ROUTINE	0.020 AC		27 QU	1.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.022 AC		45 AC	1.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.022 AC		36 AC	1.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	18	21157	64224	4	129	1676	ND	1	ND	152	87209
06/12/2017	20	15791	63530	5	275	2783	ND	4	ND	304	82408
28/06/2018	32	22904	67139	4	249	3112	ND	5	ND	290	93445

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- NO
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
CARBON DIOXIDE GAS LEVEL IS HIGH.OTHER GASES ARE ASSESSED AS ACCEPTABLE.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
30/08/2017	34	29	2.38	C
29/09/2017	40	31	1.71	B
30/10/2017	29	21	2.20	C
27/11/2017	29	20	2.10	C
06/12/2017	34	32	2.62	D
22/05/2018	33	32	2.62	D
28/06/2018	28	16	1.89	B
24/07/2018	33	30	2.46	C
25/08/2018	28	16	1.89	B
25/09/2018	30	19	2.00	B

FIELD SERVICE

DATE SERVICE
01/07/2018 PURIFIED

DATE PRINTED 10/10/2018

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **ASH HANDLING-2**LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820107
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.6%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- NO
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	3	ND	ND	3
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	ND
06/12/2017	ND	0.01	0.01
28/06/2018	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.350 AC
06/12/2017	0.000 AC	0.270 AC
28/06/2018	0.000 AC	0.490 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.22 AC
06/12/2017	0.24 AC
28/06/2018	0.20 AC

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA
06/12/2017	NON-COR	1A	LIGHT ORANGE
28/06/2018	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE SERVICE
01/07/2018 PURIFIED

No.	Unit No.	Serial Number	TC
14.	TRAFO 1 ESP UNIT 2	10037	17

A) Analysis

- **Acidity** of 0.028mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 74kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 10ppm is **ACCEPTABLE** while moisture in solid insulation of 1.05% M/dw is graded as **B**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes.**

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DATE PRINTED 10/10/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
28/11/2017	OFFLINE						
06/12/2017	OFFLINE						
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	OFFLINE						
24/07/2018	OFFLINE						
25/08/2018	OFFLINE						
25/09/2018	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10037
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
28/11/2017	ROUTINE	0.020 AC		48 AC	1.0 AC		CLEAR AC	
06/12/2017	ROUTINE	0.022 AC		22 UN	1.0 AC		CLEAR AC	
21/05/2018	ROUTINE	0.025 AC		24 UN	1.0 AC		CLEAR AC	
28/06/2018	ROUTINE	0.025 AC	32.9 AC	77 AC	1.0 AC	0.8693 AC	CLEAR AC	
24/07/2018	ROUTINE	0.028 AC		69 AC	1.0 AC		CLEAR AC	
25/08/2018	ROUTINE	0.031 AC		74 AC	1.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.028 AC		74 AC	1.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
28/06/2018	93	17225	43231	5	57	775	ND	2	ND	157	61388

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
28/11/2017	27	21	2.48	D
06/12/2017	27	35	4.13	E
21/05/2018	31	32	3.04	E
28/06/2018	28	11	1.30	B
24/07/2018	31	11	1.04	B
25/08/2018	29	12	1.26	B
25/09/2018	30	10	1.05	B

FIELD SERVICE
DATE SERVICE

DATE PRINTED 10/10/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10037
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
28/06/2018	ND	0.06	0.03

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
28/06/2018	0.030 AC	0.420 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
28/06/2018	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
28/06/2018	NON-COR	2C	LAVENDER BLUE

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
15.	TRAFO 2 ESP UNIT 2	10036	18

A) Analysis

- **Acidity** of 0.056mgKOH/gm is **QUESTIONABLE**. The levels remained stable without significant changes. Stabilized condition may proceed with annual monitoring.
- **Dielectric** of 51kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 9ppm is **ACCEPTABLE** while moisture in solid insulation of 0.94% M/dw is graded as **B**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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DATE PRINTED 10/10/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
28/11/2017	OFFLINE						
06/12/2017	OFFLINE						
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE						
24/07/2018	OFFLINE						
25/08/2018	OFFLINE						
25/09/2018	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10036
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
28/11/2017	ROUTINE	0.048 AC		55 AC	1.0 AC		CLEAR AC	
06/12/2017	ROUTINE	0.051 QU		59 AC	1.0 AC		CLEAR AC	
21/05/2018	ROUTINE	0.053 QU		55 AC	1.0 AC		CLEAR AC	
28/06/2018	ROUTINE	0.056 QU	31.3 QU	66 AC	1.0 AC	0.8573 AC	CLEAR AC	
24/07/2018	ROUTINE	0.059 QU		79 AC	1.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.062 QU		67 AC	1.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.056 QU		51 AC	1.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
28/06/2018	8	22714	64906	5	67	879	ND	ND	ND	80	88579

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
28/11/2017	28	13	1.53	C
06/12/2017	28	14	1.65	C
21/05/2018	31	16	1.52	C
28/06/2018	27	13	1.53	C
24/07/2018	31	8	0.76	A
25/08/2018	31	10	0.95	B
25/09/2018	29	9	0.94	B

FIELD SERVICE
DATE SERVICE

DATE PRINTED 10/10/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10036
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
28/06/2018	ND	42	ND	ND	ND	42

RECOMMENDATION:
FURAN ANALYSIS SHOWS MINOR DETECTION OF FURANIC COMPOUNDS. THIS IS IN LINE WITH A
HEALTHY SOLID INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
28/06/2018	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
28/06/2018	0.000 AC	0.680 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
28/06/2018	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
28/06/2018	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
16.	TRAFO 3 ESP UNIT 2	10033	19

A) Analysis

- **Acidity** of 0.067mgKOH/g is **QUESTIONABLE**. The levels remained stable without further oil degradation. Stabilized condition may proceed with annual monitoring.
- **Dielectric** of 59kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 12ppm is **ACCEPTABLE** while moisture in solid insulation of 1.26% M/dw is graded as **B**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- Based on previous oil analysis, the IFT of the oil is unfavorable. Together with the high level of Acidity, it is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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DATE PRINTED 10/10/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
28/11/2017	OFFLINE						
06/12/2017	OFFLINE						
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE						
24/07/2018	OFFLINE						
25/08/2018	OFFLINE						
25/09/2018	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10033
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
28/11/2017	ROUTINE	0.053 QU		42 AC	1.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.056 QU		60 AC	1.5 AC		CLEAR AC	
21/05/2018	ROUTINE	0.059 QU		51 AC	1.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.062 QU	26.3 UN	78 AC	1.5 AC	0.8602 AC	CLEAR AC	
24/07/2018	ROUTINE	0.070 QU		67 AC	1.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.067 QU		73 AC	1.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.067 QU		59 AC	1.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
28/06/2018	14	17627	48672	4	47	935	ND	ND	ND	65	67299

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
28/11/2017	28	15	1.77	D
06/12/2017	28	16	1.89	D
21/05/2018	31	18	1.71	D
28/06/2018	28	14	1.65	C
24/07/2018	32	12	1.14	B
25/08/2018	30	13	1.37	C
25/09/2018	30	12	1.26	B

FIELD SERVICE
DATE SERVICE

DATE PRINTED 10/10/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10033
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
28/06/2018	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
28/06/2018	0.000 AC	0.570 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
28/06/2018	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
28/06/2018	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
17.	TRAFO 4 ESP UNIT 2	10032	20

A) Analysis

- **Acidity** of 0.031mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 51kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 11ppm is **ACCEPTABLE** while moisture in solid insulation of 1.15% M/dw is graded as **B**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

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DATE PRINTED 10/10/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
28/11/2017	OFFLINE						
06/12/2017	OFFLINE						
21/05/2018	OFFLINE					GOOD	NONE
28/06/2018	ONLINE						
24/07/2018	OFFLINE						
25/08/2018	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10032
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
28/11/2017	ROUTINE	0.020 AC		62 AC	1.0 AC		CLEAR AC	
06/12/2017	ROUTINE	0.022 AC		58 AC	1.0 AC		CLEAR AC	
21/05/2018	ROUTINE	0.025 AC		40 AC	1.0 AC		CLEAR AC	
28/06/2018	ROUTINE	0.028 AC	32.6 AC	88 AC	1.0 AC	0.8648 AC	CLEAR AC	
24/07/2018	ROUTINE	0.031 AC		59 AC	1.0 AC		CLEAR AC	
25/08/2018	ROUTINE	0.031 AC		68 AC	1.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.031 AC		51 AC	1.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
28/06/2018	22	25201	74464	5	85	823	ND	ND	ND	112	100600

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
28/11/2017	28	16	1.89	D
06/12/2017	28	17	2.01	D
21/05/2018	31	18	1.71	D
28/06/2018	27	15	1.77	D
24/07/2018	32	13	1.24	B
25/08/2018	31	10	0.95	B
25/09/2018	29	11	1.15	B

FIELD SERVICE
DATE SERVICE

DATE PRINTED 10/10/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10032
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
28/06/2018	ND	ND	0.03

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
28/06/2018	0.000 AC	1.620 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
28/06/2018	0.01 UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
28/06/2018	NON-COR	2C	LAVENDER BLUE

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
18.	MAIN TRAFO UNIT 2	1006902	21

A) Analysis

- **Acidity** of 0.011mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 90kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 7ppm is **ACCEPTABLE** while moisture in solid insulation of 0.67% M/dw is graded as **A**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

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DATE PRINTED 10/10/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAFU UNIT 2

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
27/11/2017	ONLINE	50	55				
06/12/2017	ONLINE						
22/05/2018	ONLINE			NORMAL		GOOD	NONE
28/06/2018	OFFLINE	50	45				
24/07/2018	ONLINE					GOOD	
25/08/2018	ONLINE						
25/09/2018	ONLINE						

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 1006902
KVA	- 33000
PRI	- 150000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.03%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30857

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 SEDIMENT
27/11/2017	ROUTINE	0.008 AC		70 AC	0.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.008 AC		67 AC	0.5 AC		CLEAR AC	
22/05/2018	ROUTINE	0.011 AC		46 AC	0.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.008 AC	42.1 AC	75 AC	0.5 AC	0.8663 AC	CLEAR AC	
24/07/2018	ROUTINE	0.011 AC		79 AC	0.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.011 AC		83 AC	0.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.011 AC		90 AC	0.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
28/06/2018	19	12733	53386	10	193	2448	ND	2	ND	224	68791

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
27/11/2017	37	16	1.04	B
06/12/2017	38	13	0.85	A
22/05/2018	33	14	1.15	B
28/06/2018	28	10	1.18	B
24/07/2018	33	11	0.90	B
25/08/2018	34	9	0.74	A
25/09/2018	32	7	0.67	A

FIELD SERVICE
DATE SERVICE

DATE PRINTED 10/10/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAF0 UNIT 2

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 1006902
KVA	- 33000
PRI	- 150000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.03%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30857
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	113	113

RECOMMENDATION:
RETEST 6 MONTHS: THE PRESENCE OF 5M2F INDICATES THE POSSIBILITY OF HOTSPOT
OVERHEATING.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
28/06/2018	ND	ND	0.03

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
28/06/2018	0.000 AC	0.410 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
28/06/2018	0.22 AC

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
28/06/2018	NON-COR	1B	DARK ORANGE

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

3.0 Summary of Our Comments & Recommendations:

No.	Sub Name	Unit No.	TC	Recommendation
1.	MAIN TRAFU UNIT 1	100691	4	• Periodic condition monitoring
2.	TRAFU 1 ESP UNIT 1	10937	5	• Periodic condition monitoring
3.	TRAFU 2 ESP UNIT 1	10035	6	• Periodic condition monitoring
4.	TRAFU 3 ESP UNIT 1	10031	7	• Perform OIL REGENERATION
5.	TRAFU 4 ESP UNIT 1	10034	8	• Perform OIL REGENERATION
6.	MAIN TRAFU UNIT 3	1122170	9	• Periodic condition monitoring
7.	UAT 3	1035082	10	• Periodic condition monitoring
8.	TRAFU 1 ESP UNIT 3	0073084208	11	• Periodic condition monitoring
9.	TRAFU 2 ESP UNIT 3	0070084208	12	• Perform OIL REGENERATION
10.	TRAFU 3 ESP UNIT 3	0072084208	13	• Perform OIL PURIFICATION
11.	TRAFU 4 UNIT 3	0071084208	14	• Periodic condition monitoring
12.	ASH HANDLING-1	10820108	15	• Periodic condition monitoring
13.	ASH HANDLING-2	10820107	16	• Periodic condition monitoring
14.	TRAFU 1 ESP UNIT 2	10037	17	• Periodic condition monitoring
15.	TRAFU 2 ESP UNIT 2	10036	18	• Periodic condition monitoring
16.	TRAFU 3 ESP UNIT 2	10033	19	• Perform OIL REGENERATION

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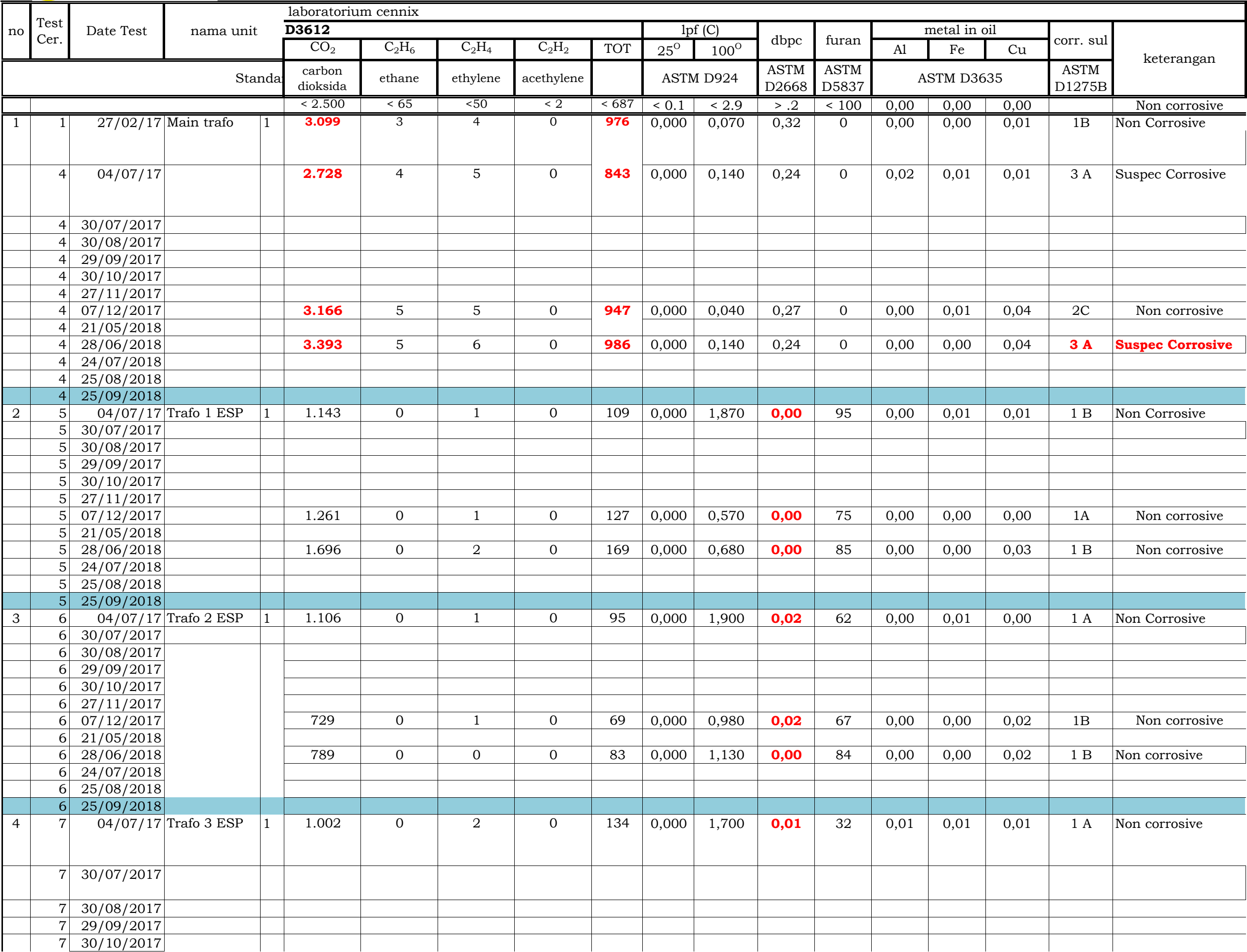
No.	Sub Name	Unit No.	TC	Recommendation
17.	TRAFO 4 ESP UNIT 2	10032	20	• Periodic condition monitoring
18.	MAIN TRAFO UNIT 2	1006902	21	• Periodic condition monitoring

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no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil test														
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture		dga ASTM					
															ppm	grade	H ₂	O ₂	N ₂	CH ₄	CO	
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533	hydrogen	oksigen	nitrogen	methane	carbon monoksida		
							BATASAN OPERASI	< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20		< 100			< 120	< 350	
1	1	27/02/17	Main trafo	1	1006901	2011	30.857	Wolong	0,014	39,9	99	0,5	0,8693	C		9	A	135	15.834	57.625	27	807
	4	04/07/17							0,017	41,5	58	0,5	0,8691	C		10	A	147	9.471	4.328	32	655
	4	30/07/2017							0,017		70	0,5		C		8	B					
	4	30/08/2017							0,017		92	0,5		C		6	A					
	4	29/09/2017							0,014		82	0,5		C		8	A					
	4	30/10/2017							0,014		82	0,5		C		6	A					
	4	27/11/2017							0,014		53	0,5		C		8	A					
	4	07/12/2017							0,011	40,1	87	0,5	0,8745	C		7	A	183	12.895	43.794	39	715
	4	21/05/2018							0,011		77	0,5		C		10	B					
	4	28/06/2018							0,014	41,9	69	0,5	0,8716	C		7	A	176	11.044	46.776	41	758
	4	24/07/2018							0,014		86	0,5		C		6	A					
	4	25/08/2018							0,017		90	0,5		C		6	A					
	4	25/09/2018							0,014		96	0,5				7	A					
2	5	04/07/17	Trafo 1 ESP	1	10937	2010	1.143	Fujia	0,056	25,3	50	1,5	0,8529	C		18	D	35	20.494	67.487	4	69
	5	30/07/2017							0,059		57	1,5		C		20	D					
	5	30/08/2017							0,056		65	1,5		C		18	D					
	5	29/09/2017							0,059		58	1,5		C		19	D					
	5	30/10/2017							0,059		23	1,5		C		32	C					
	5	27/11/2017							0,062		25	1,5		C		33	C					
	5	07/12/2017							0,062	25,0	49	1,5	0,8593	C		18	D	39	19.933	67.198	4	83
	5	21/05/2018							0,065		38	1,5		C		33	E					
	5	28/06/2018							0,067	25,9	32	1,5	0,8560	C		30	E	44	20.039	64.529	4	119
	5	24/07/2018							0,073		38	1,5		C		19	D					
	5	25/08/2018							0,020		71	0,5		C		12	B					
	5	25/09/2018							0,017		63	0,5				18	B					
3	6	04/07/17	Trafo 2 ESP	1	10035	2010	1.143	Fujia	0,042	26,6	41	1,5	0,8540	C		21	D	27	18.639	66.781	3	64
	6	30/07/2017							0,045		25	1,5		C		27	E					
	6	30/08/2017							0,042		42	1,5		C		14	C					
	6	29/09/2017							0,048		59	1,5		C		19	D					
	6	30/10/2017							0,051		62	1,5		C		17	B					
	6	27/11/2017							0,051		50	1,5		C		18	B					
	6	07/12/2017							0,063	27,3	51	1,5	0,8589	C		17	D	20	19.360	61.590	3	45
	6	21/05/2018							0,056		45	1,5		C		20	D					
	6	28/06/2018							0,059	27,6	55	1,5	0,8570	C		17	C	22	22.494	64.443	4	57
	6	24/07/2018							0,062		47	1,5		C		15	C					
	6	25/08/2018							0,022		56	0,5		C		13	B					
	6	25/09/2018							0,020		67	0,5				11	B					
4	7	04/07/17	Trafo 3 ESP	1	10031	2010	1.143	Fujia	0,048	28,4	44	1,0	0,8537	C		32	E	32	15.273	65.111	5	95
	7	30/07/2017							0,042		77	1,5				10	D					
	7	30/08/2017							0,042		61	1,5		C		13	C					
	7	29/09/2017							0,045		72	1,5		C		14	B					
	7	30/10/2017							0,048		72	1,5		C		16	B					





no	Test Cer.	Date Test	nama unit	MALAYSIAN LAB RECOMMENDATION		PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION
Standar						
1	1	27/02/17	Main trafo	1	Retest 6 months: DGA Check leakages Other parameters: Periodic Condition Monitoring	For TEST 25 September 2018 PURIFIKASI Pada trafo ESP 4 Unit 1 dan ESP 3 Unit 3 kemungkinan pekerjaan Purifikasi/Ganti Minyak dilakukan kurang sempurna sehingga terjadi penurunan BDV dan Moisture dengan cepat. Kami sarankan untuk melakukan Purifikasi dengan temperatur lebih dari 650C dan vacuum diatas 74 CMHg diatas 7 cycle. GANTI MINYAK Kami sarankan untuk trafo ESP 3 Unit 1, ESP 2 Unit 3 dan ESP 2 dan 3 pada Unit 2 dilakukan penggantian minyak. seperti pada trafo ESP 2 unti 3, meskipun telah dilakukan Purifikasi, ternyata terjadi penurunan BDV dan Moisture dengan relatif cepat. Kami sarankan pada saat pengisian kembali dilakukan dengan cara purifikasi dengan temperatur lebih dari 650C dan vacuum diatas 74 CMHg diatas 7 cycle.
	4	04/07/17			Retest 6 months: DGA Add inhibitor 2,6-DBPC and Passivator Other parameters: Periodic Condition Monitoring	
	4	30/07/2017			Periodic condition monitoring	
	4	30/08/2017			Periodic condition monitoring	
	4	29/09/2017			Periodic condition monitoring	For TEST 25 Agustus 2018 Traf di Unit 1 dan 3 pada saat ini dalam kondisi yang optimal untuk operasional setelah dilakukan Purifikasi dan Penggantian Oli. Untuk Trafo ESP 2 dan 3 pada Unit 2 , kami sarankan untuk dilakukan ganti oli mengingat trend penurunan pada nilai Acidity. For TEST 24 Juli 2018 GANTI MINYAK Disarankan dilakukan untuk beberapa trafo ESP (Unit 1 no 1 dan 2 , Unit 2 no 3) karena nilai Acid sudah melewati standard. Sedang untuk ESP Unit 2 no 2 ada trend Acid yang memburuk, dipersiapkan untuk penggantian oli juga. PURIFIKASI Disarankan dilakukan pada trafo Ash Handling no 2 , Trafo ESP Unit 1 no 3 dan 4 , karena Water Content Grade di bawah standard. Purifikasi agar dilakukan dengan temperatur tinggi dan lebih dari 5 cycle, agar berpengaruh ke Water Content Grade FOR TEST 28 Juni 2018 Masih ada beberapa trafo ESP (no 1 Unit 1 dan no 2 Unit 2) yang Acid dan IFT di bawah standard. Kami sarankan untuk ganti minyak. Trafo-trafo ESP (no 2,3,4 unit 1 , no 2,3,4 Unit 2 , n 2,3 Unit 3) dan Ash Handling 1 yang Grade Mixture di bawah B, kami sarankan untuk dilakukan Purifikasi
	4	30/10/2017			Periodic condition monitoring	
	4	27/11/2017			Periodic condition monitoring	
	4	07/12/2017			Retest 6 mnth : DGA	
	4	21/05/2018			Periodic condition monitoring	
	4	28/06/2018			Add passivator	
	4	24/07/2018			Periodic condition monitoring	
	4	25/08/2018			Periodic condition monitoring	
	4	25/09/2018			Periodic condition monitoring	
2	5	04/07/17	Trafo 1 ESP	1	Perform OIL REGENERATION	
	5	30/07/2017			Perform OIL REGENERATION	
	5	30/08/2017			Perform OIL REGENERATION	
	5	29/09/2017			Perform OIL REGENERATION	
	5	30/10/2017			Perform OIL REGENERATION	
	5	27/11/2017			Perform OIL REGENERATION	
	5	07/12/2017			Perform OIL REGENERATION	
	5	21/05/2018			Perform OIL REGENERATION	
	5	28/06/2018			Perform OIL REGENERATION	
	5	24/07/2018			Perform OIL REGENERATION	
	5	25/08/2018			Periodic condition monitoring (change oil on beginning of August 2018)	
	5	25/09/2018			Periodic condition monitoring	
3	6	04/07/17	Trafo 2 ESP	1	Perform OIL REGENERATION	FOR TEST 28 Juni 2018 Masih ada beberapa trafo ESP (no 1 Unit 1 dan no 2 Unit 2) yang Acid dan IFT di bawah standard. Kami sarankan untuk ganti minyak. Trafo-trafo ESP (no 2,3,4 unit 1 , no 2,3,4 Unit 2 , n 2,3 Unit 3) dan Ash Handling 1 yang Grade Mixture di bawah B, kami sarankan untuk dilakukan Purifikasi
	6	30/07/2017			Perform OIL REGENERATION	
	6	30/08/2017			Perform OIL REGENERATION	
	6	29/09/2017			Perform OIL REGENERATION	
	6	30/10/2017			Perform OIL REGENERATION	
	6	27/11/2017			Perform OIL REGENERATION	
	6	07/12/2017			Perform OIL REGENERATION	
	6	21/05/2018			Perform OIL REGENERATION	
	6	28/06/2018			Perform OIL REGENERATION	
	6	24/07/2018			Perform OIL REGENERATION	
	6	25/08/2018			Periodic condition monitoring (change oil on beginning of August 2018)	
	6	25/09/2018			Periodic condition monitoring	
4	7	04/07/17	Trafo 3 ESP	1	Retest 6 months: Acidity, IFT, LPF, Diel and Water Content Add inhibitor 2,6-DBPC and Passivator Other parameters: Periodic Condition Monitoring	
	7	30/07/2017			Retest 6 months: Acidity, IFT and Liquid Power Factor Other parameters: Periodic Condition Monitoring	
	7	30/08/2017			Periodic condition monitoring	
	7	29/09/2017			Periodic condition monitoring	
	7	30/10/2017			Periodic condition monitoring	



no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil test														
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture		dga ASTM					
															ppm	grade	H ₂	O ₂	N ₂	CH ₄	CO	
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533	hydrogen	oksigen	nitrogen	methane	carbon monoksida		
		BATASAN OPERASI						< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20		< 100			< 120	< 350	
	7	27/11/2017						0,051		62	1,5		C		17	C						
	7	07/12/2017						0,063	29,3	57	1,5	0,8624	C		16	C	7	14.730	65.738	3	42	
	7	21/05/2018						0,056		48	1,5		C		21	D						
	7	28/06/2018						0,053	29,7	62	1,5	0,8605	C		17	D	9	17.958	66.007	4	46	
	7	24/07/2018						0,510		64	1,5		C		17	D						
	7	25/08/2018						0,056		57	1,5		C		15	C						
	7	25/09/2018						0,051		46	1,5				21	D						
5	8	04/07/17	Trafo 4 ESP	1	10034	2010	1.143	Fujia	0,028	26,3	24	1,5	0,8596	C		29	E	13	12.265	66.770	5	172
	8	30/07/2017						0,022		88	1,5		C		15	D						
	8	30/08/2017						0,028		44	1,5		C		13	C						
	8	29/09/2017						0,031		53	1,0		C		15	B						
	8	30/10/2017						0,028		48	1,0		C		18	B						
	8	27/11/2017						0,034		41	1,5		C		30	D						
	8	07/12/2017						0,031	27,2	41	1,5	0,8645	C		18	D	9	13.331	63.527	3	70	
	8	21/05/2018						0,034		37	1,5		C		23	D						
	8	28/06/2018						0,037	27,5	55	1,5	0,8653	C		20	D	9	17.962	64.248	4	75	
	8	24/07/2018						0,039		50	1,5		C		19	D						
	8	25/08/2018						0,017		61	0,5		C		14	C						
	8	25/09/2018						0,014		45	0,5				20	D						
6	2	27/02/17	Main trafo	3	1122170	2010	26.706	Shandong	0,017	39,7	87	0,5	0,8781	C		9	A	4	13.330	43.140	9	808
	9	04/07/17						0,017	42,3	78	0,5	0,8780	C		15	B	24	10.131	44.084	11	641	
	9	30/07/2017						0,014		77	0,5		C		17	B						
	9	30/08/2017						0,017		86	0,5		C		8	A						
	9	29/09/2017						0,014		86	0,5		C		10	A						
	9	30/10/2017						0,014		88	0,5		C		9	A						
	9	27/11/2017						0,014		87	0,5		C		8	A						
	9	07/12/2017						0,014	41,5	84	0,5	0,8932	C			A	39	14.023	43.226	11	668	
	9	21/05/2018						0,014		58	0,5		C		11	B						
	9	28/06/2018						0,014	41,0	85	0,5	0,8805	C		8	A	18	11.431	48.283	13	700	
	9	24/07/2018						0,014		77	0,5		C		8	A						
	9	25/08/2018						0,017		82	0,5		C		7	A						
	9	25/09/2018						0,017		83	0,5				6	A						
7	3	27/02/17	Trafo UAT	3	1035082	2010	4.749	Shandong	0,017	37,6	94	0,5	0,8768	C		9	A	20	12.535	56.879	20	985
	10	04/07/17						0,022	38,1	60	0,5	0,8767	C		11	A	20	16.028	68.308	20	971	
	10	30/07/2017						0,025		88	0,5		C		14	B						
	10	30/08/2017						0,022		89	0,5		C		11	A						
	10	29/09/2017						0,020		80	0,5		C		12	A						
	10	30/10/2017						0,022		86	0,5		C		11	A						
	10	27/11/2017						0,022		86	0,5		C		12	A						
	10	07/12/2017						0,020	37,6	86	0,5	0,8200	C		12	A	36	18.752	66.758	1	1.116	
	10	21/05/2018						0,020		81	0,5		C		14	B						
	10	28/06/2018						0,022	37,5	79	0,5	0,8793	C		10	A	35	13.798	68.180	22	1.144	



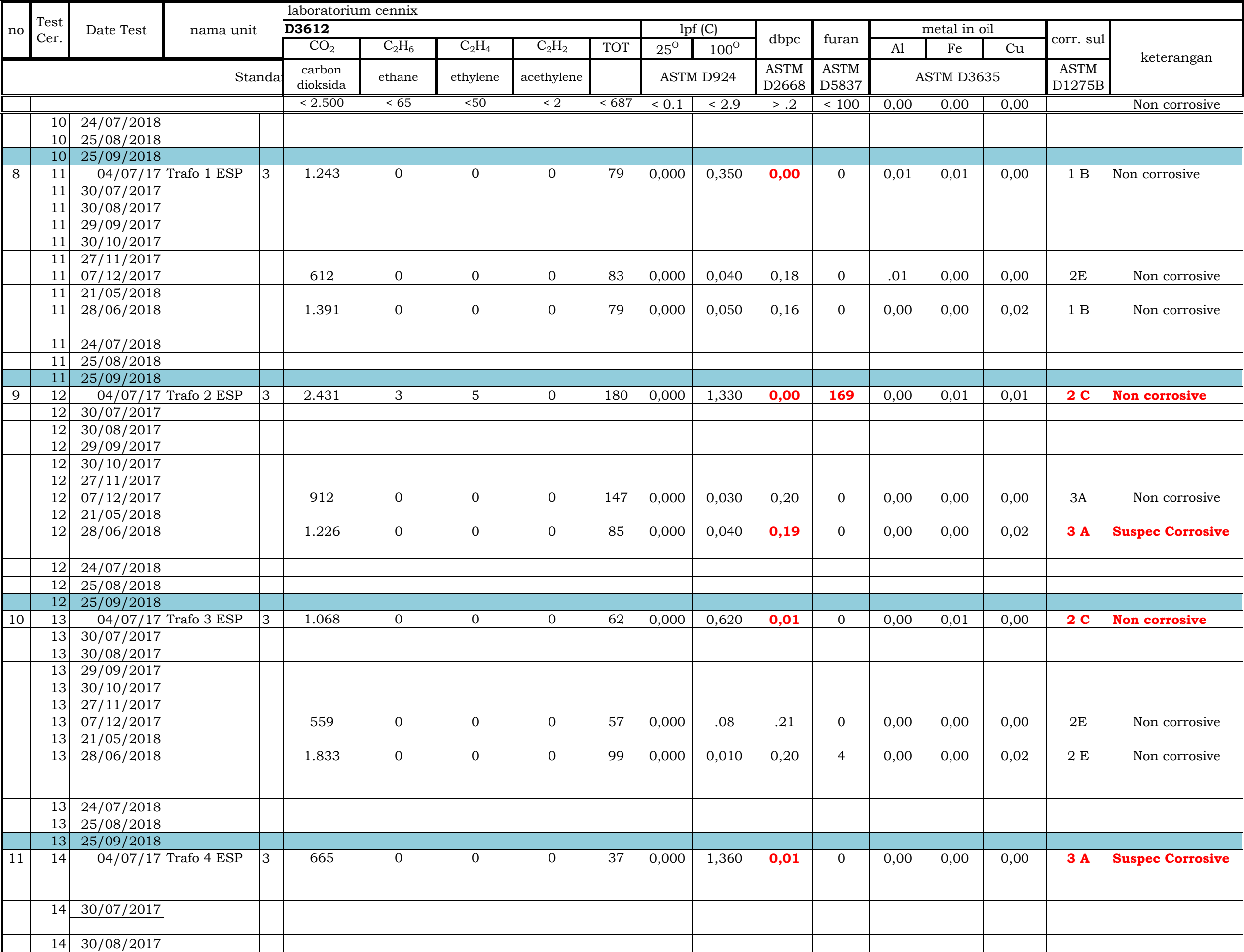
no	Test Cer.	Date Test	nama unit		laboratorium cennix													keterangan
					D3612					lpf (C)		dbpc	furan	metal in oil			corr. sul	
					CO ₂	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	TOT	25 ^o	100 ^o			Al	Fe	Cu		
Standar				carbon dioksida	ethane	ethylene	acethylene		ASTM D924		ASTM D2668	ASTM D5837	ASTM D3635			ASTM D1275B		
					< 2.500	< 65	<50	< 2	< 687	< 0.1	< 2.9	> .2	< 100	0,00	0,00	0,00		Non corrosive
	7	27/11/2017																
	7	07/12/2017			617	0	0	0	52	0,000	0,630	0,01	14	0,00	0,00	0,00	2C	Non corrosive
	7	21/05/2018																
	7	28/06/2018			781	0	1	0	60	0,000	1,030	0,00	20	0,00	0,00	0,11	1 B	Non corrosive
	7	24/07/2018																
	7	25/08/2018																
	7	25/09/2018																
5	8	04/07/17	Trafo 4 ESP	1	1.206	0	8	0	198	0,180	6,680	0,08	18	0,05	0,02	0,01	2 C	Non Corrosive
	8	30/07/2017																
	8	30/08/2017																
	8	29/09/2017																
	8	30/10/2017																
	8	27/11/2017																
	8	07/12/2017			720	0	1	0	83	0,010	2,180	0,06	0	0,00	0,00	0,02	2C	Non corrosive
	8	21/05/2018																
	8	28/06/2018			1.054	0	2	0	90	0,060	2,850	0,02	21	0,00	0,00	0,05	3 A	Suspec Corrosive
	8	24/07/2018																
	8	25/08/2018																
	8	25/09/2018																
6	2	27/02/17	Main trafo	3	2.878	0	1	0	822	0,000	0,090	0,30	0	0,00	0,03	0,02	1A	Non corrosive
	9	04/07/17			2.250	1	1	0	678	0,000	0,130	0,22	0	0,01	0,01	0,02	3 A	Suspec Corrosive
	9	30/07/2017																
	9	30/08/2017																
	9	29/09/2017																
	9	30/10/2017																
	9	27/11/2017																
	9	07/12/2017			2.456	0	1	0	719	0,000	0,070	0,22	0	0,01	0,00	0,03	2C	Non corrosive
	9	21/05/2018																
	9	28/06/2018			1.512	1	1	0	733	0,000	0,110	0,19	0	0,00	0,01	0,06	3 A	Suspec Corrosive
	9	24/07/2018																
	9	25/08/2018																
	9	25/09/2018																
7	3	27/02/17	Trafo UAT	3	3.877	6	7	0	1.038	0,000	0,100	0,30	0	0,00	0,01	0,04	1A	Non Corrosive
	10	04/07/17			3.536	5	7	0	1.023	0,000	0,390	0,23	0	0,02	0,01	0,04	3 A	Suspec Corrosive
	10	30/07/2017																
	10	30/08/2017																
	10	29/09/2017																
	10	30/10/2017																
	10	27/11/2017																
	10	07/12/2017			3.622	5	7	0	1.165	0,000	0,170	0,23	0	0,00	0,01	0,08	2C	Non corrosive
	10	21/05/2018																
	10	28/06/2018			3.644	5	7	1	1.214	0,000	0,140	0,21	0	0,00	0,01	0,11	3 A	Suspec Corrosive



no	Test Cer.	Date Test	nama unit	MALAYSIAN LAB RECOMMENDATION		PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION
Standar						
	7	27/11/2017			Periodic condition monitoring	Pada ESP 2,3,4 Unit 3 ditemukan Spesific Grafity di bawah stadanrd. Kemungkinan penggantian oli menggunakan bukan Mineral Oli. Mohon dicek kembali type oli pengganti atau dilakukan trending dalam 12 bulan mendatang. FURAN Semua Trafo ESP Unit 1, ESP 2 Unit 2 dan ESP 3 Unit 3 telah ditemukan jejak Furan akibat moisture tinggi yang cukup lama. Kami sarankan untuk menjaga kondisi moisture pada waktu selanjutnya agar perkembangan Furan tidak berlanjut Main Trafo Unit 2 ditemukan jejak Furan 5M2F (ada hot spot) tetapi tidak ditemukan Combustible Gas berlebih pada test DGA. Hal ini mengakibatkan pengamanan terhadapat gangguan petir kurang memadai. Kami sarankan untuk diperbaiki baiksystem groundingnya maupun arresternya CORROSIVE SULPHUR Main Trafo Unit 1 dan 3 serta trafo UAT dideteksi mulai ada corrosive sulphur. Untuk mencegah percepatan pertumbuhannya dan memperpanjang umur trafo, kami sarankan menambah Passivator FOR TEST 21 Mei 2018 Mengingat banyak trafo ESP (Tc 5,6,7,8,11,12,13) yang kurang memenuhi syarat. Kami sarankan untuk diganti minyak saja. Selain jumlah minyaknya sedikit biaya juga akan jauh lebih murah, Saran kami saat penggantian minyak melalui purifikasi dengan temperatur lebih dari 650C dan vacuum diatas 74 CMHg diatas 7 cycle Trafo ESP (Tc 6,7,8) Furan (FOL) banyak berkurang karena moisture cukup banyak dalam kertas dan kemungkinan juga kualitas kertas kurang bagus Untuk trafo Ash Handling sebaiknya segera dipurifikasi juga dengan temperatur diats 650c dan vacuum diatas 74 CMHg sekitar >7 cycle Trafo ESP 1 unit 2, nilai Dielectric nya cukup rendah ditambah dengan kondisi Moisture yang buruk, sebaiknya segera dilakukan Purifikasi. Kondisi semua trafo ESP Unit 2 (Tc 17, 18, 19, 20) Grade Moisture rendah, yang artinya kandungan air dalam isolasi kertas cukup tinggi. Kami sarankan untuk dimonitor dengan test DGA agar bisa diantisipasi jika ada kondisi gangguan thermal yang terjadi pada trafo trafo tersebut.
	7	07/12/2017			Retest 6 months: Acidity, IFT	
	7	21/05/2018			Retest 6 months: Acidity, IFT, Dielectric and Water Content	
	7	28/06/2018			Add inhibitor 2,6-DBPC and PASSIVATOR	
	7	24/07/2018			Periodic condition monitoring	
	7	25/08/2018			Periodic condition monitoring	
	7	25/09/2018			Perform OIL REGENERATION	
5	8	04/07/17	Trafo 4 ESP	1	Perform OIL REGENERATION Inspect silica gel	
	8	30/07/2017			Perform OIL REGENERATION	
	8	30/08/2017			Perform OIL REGENERATION	
	8	29/09/2017			Perform OIL REGENERATION	Perform OIL REGENERATION Periodic condition monitoring (change oil on beginning of August 2018) Perform OIL PURIFICATION
	8	30/10/2017			Perform OIL REGENERATION	
	8	27/11/2017			Perform OIL REGENERATION	
	8	07/12/2017			Perform OIL REGENERATION	
	8	21/05/2018			Perform OIL REGENERATION	
	8	28/06/2018			Perform OIL REGENERATION	
	8	24/07/2018			Perform OIL REGENERATION	
	8	25/08/2018			Periodic condition monitoring (change oil on beginning of August 2018)	
	8	25/09/2018			Perform OIL PURIFICATION	
6	2	27/02/17	Main trafo	3	Retest 6 months: DGA	
					Other parameters: Periodic Condition Monitoring	Add Passivator Periodic Condition Monitoring Periodic condition monitoring Periodic condition monitoring Periodic condition monitoring Periodic condition monitoring Periodic condition monitoring Periodic condition monitoring Periodic condition monitoring Periodic condition monitoring Add passivator Periodic condition monitoring Periodic condition monitoring Periodic condition monitoring
	9	04/07/17			Add Passivator	
					Periodic Condition Monitoring	
	9	30/07/2017			Periodic condition monitoring	
	9	30/08/2017			Periodic condition monitoring	
	9	29/09/2017			Periodic condition monitoring	
	9	30/10/2017			Periodic condition monitoring	
	9	27/11/2017			Periodic condition monitoring	
	9	07/12/2017			Periodic condition monitoring	
	9	21/05/2018			Periodic condition monitoring	
	9	28/06/2018			Add passivator	Periodic condition monitoring Retest 6 months: DGA Other parameters: Periodic Condition Monitoring Add Passivator Periodic Condition Monitoring Inspect silica gel Periodic condition monitoring Periodic condition monitoring Periodic condition monitoring Periodic condition monitoring Periodic condition monitoring Periodic condition monitoring Periodic condition monitoring Retest 6 months: DGA Add Passivator
	9	24/07/2018			Periodic condition monitoring	
	9	25/08/2018			Periodic condition monitoring	
	9	25/09/2018			Periodic condition monitoring	
7	3	27/02/17	Trafo UAT	3	Retest 6 months: DGA	
					Other parameters: Periodic Condition Monitoring	
	10	04/07/17			Add Passivator	
					Periodic Condition Monitoring	
					Inspect silica gel	
	10	30/07/2017			Periodic condition monitoring	
	10	30/08/2017			Periodic condition monitoring	
	10	29/09/2017			Periodic condition monitoring	
	10	30/10/2017			Periodic condition monitoring	
	10	27/11/2017			Periodic condition monitoring	
	10	07/12/2017			Periodic condition monitoring	
	10	21/05/2018			Periodic condition monitoring	
	10	28/06/2018			Retest 6 months: DGA	
					Add Passivator	



no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil test														
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture		dga ASTM					
															ppm	grade	H ₂	O ₂	N ₂	CH ₄	CO	
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533	hydrogen	oksigen	nitrogen	methane	carbon monoksida		
			BATASAN OPERASI					< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20		< 100			< 120	< 350	
	10	24/07/2018						0,020		76	0,5		C		11	A						
	10	25/08/2018						0,022		78	0,5		C		8	A						
	10	25/09/2018						0,020		75	0,5				8	A						
8	11	04/07/17	Trafo 1 ESP	3	007308428	2010	731	Zhejian	0,022	23,2	32	1,5	0,8432	C		26	D	14	12.943	68.807	3	62
	11	30/07/2017						0,028		47	1,5		C		21	D						
	11	30/08/2017						0,028		36	1,5		C		22	D						
	11	29/09/2017						0,025		45	1,5		C		20	C						
	11	30/10/2017						0,025		40	1,5		C		19	D						
	11	27/11/2017						0,025		31	1,5		C		34	C						
	11	07/12/2017						0,022	34,6	31	1,5	0,8190	C		31	C	14	12.739	65.097	3	66	
	11	21/05/2018						0,022		31	0,5		C		27	D						
	11	28/06/2018						0,022	35,0	60	0,5	0,8154	C		10	B	19	15.552	64.448	3	57	
	11	24/07/2018						0,020		53	0,5		C		9	B						
	11	25/08/2018						0,022		77	0,5		C		9	B						
	11	25/09/2018						0,020		64	0,5				10	B						
9	12	04/07/17	Trafo 2 ESP	3	007008428	2010	731	Zhejian	0,118	16,2	24	3,0	0,8435	C		39	D	38	16.781	70.523	5	129
	12	30/07/2017						0,112		28	3,0		C		38	E						
	12	30/08/2017						0,109		34	2,5		C		17	D						
	12	29/09/2017						0,115		21	2,5		C		37	D						
	12	30/10/2017						0,118		33	2,5		C		20	D						
	12	27/11/2017						0,017		25	0,5		C		28	B						
	12	07/12/2017						0,022	29,7	29	0,1	0,8106	C		44	D	42	19.523	66.059	3	102	
	12	21/05/2018						0,022		28	0,5		C		30	E						
	12	28/06/2018						0,025	30,2	34	0,5	0,8080	C		16	C	43	22.272	68.719	4	38	
	12	24/07/2018						0,022		30	0,5		C		25	D						
	12	25/08/2018						0,022		67	0,5		C		11	C						
	12	25/09/2018						0,022		31	0,5				32	D						
10	13	04/07/17	Trafo 3 ESP	3	007208428	2010	731	Zhejian	0,025	27,8	40	1,5	0,8415	C		27	D	21	16.364	70.694	3	38
	13	30/07/2017						0,025		66	1,5		C		18	D						
	13	30/08/2017						0,025		50	1,5		C		21	C						
	13	29/09/2017						0,028		59	1,5		C		22	C						
	13	30/10/2017						0,028		55	1,5		C		28	D						
	13	27/11/2017						0,014		39	0,5		C		24	B						
	13	07/12/2017						0,014	39,8	45	0,5	0,8099	C		29	B	15	14.515	65.277	2	40	
	13	21/05/2018						0,014		54	0,5		C		23	D						
	13	28/06/2018						0,017	38,4	43	0,5	0,8075	C		22	C	30	15.202	64.188	4	65	
	13	24/07/2018						0,014		43	0,5		C		20	C						
	13	25/08/2018						0,014		76	0,5		C		11	C						
	13	25/09/2018						0,020		47	0,5				20	B						
11	14	04/07/17	Trafo 4 ESP	3	007108428	2010	731	Zhejian	0,020	31,2	62	1,0	0,8420	C		19	D	10	18.669	66.036	3	24
	14	30/07/2017						0,022		60	1,0		C		15	C						
	14	30/08/2017						0,020		55	1,0		C		17	D						

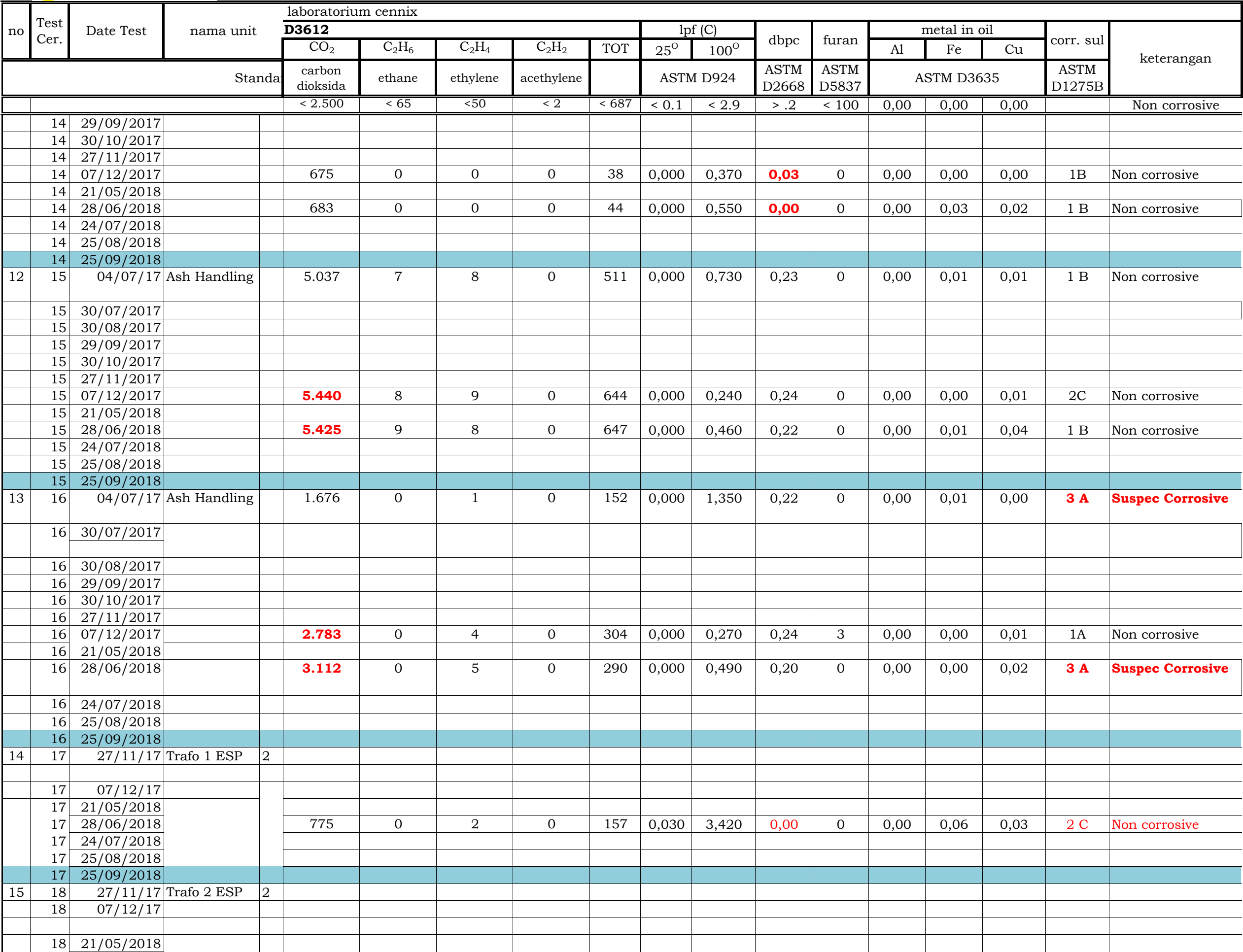




no	Test Cer.	Date Test	nama unit	MALAYSIAN LAB RECOMMENDATION		PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION
Standards						
	10	24/07/2018			Periodic condition monitoring	
	10	25/08/2018			Periodic condition monitoring	
	10	25/09/2018			Periodic condition monitoring	
8	11	04/07/17	Trafo 1 ESP	3	Perform OIL REGENERATION	
	11	30/07/2017			Perform OIL REGENERATION	
	11	30/08/2017			Perform OIL REGENERATION	
	11	29/09/2017			Perform OIL REGENERATION	
	11	30/10/2017			Perform OIL REGENERATION	
	11	27/11/2017			Perform OIL REGENERATION	
	11	07/12/2017			Perform OIL PURIFICATION	
	11	21/05/2018			Perform OIL PURIFICATION	
	11	28/06/2018			Retest 12 months: Specific Gravity Add Passivator	
	11	24/07/2018			Periodic condition monitoring	
	11	25/08/2018			Periodic condition monitoring (Purifying on end of July 2018)	
	11	25/09/2018			Periodic condition monitoring	
9	12	04/07/17	Trafo 2 ESP	3	Perform OIL REGENERATION	
	12	30/07/2017			Perform OIL REGENERATION	
	12	30/08/2017			Perform OIL REGENERATION	
	12	29/09/2017			Perform OIL REGENERATION	
	12	30/10/2017			Perform OIL REGENERATION	
	12	27/11/2017			Perform OIL REGENERATION	
	12	07/12/2017			Perform OIL REGENERATION	
	12	21/05/2018			Perform OIL PURIFICATION	
	12	28/06/2018			Retest 12 months: Specific Gravity Add Passivator	
	12	24/07/2018			Perform OIL REGENERATION	
	12	25/08/2018			Periodic condition monitoring (Purifying on end of July 2018)	
	12	25/09/2018			Perform OIL REGENERATION	
10	13	04/07/17	Trafo 3 ESP	3	Perform OIL REGENERATION	
	13	30/07/2017			Perform OIL REGENERATION	
	13	30/08/2017			Perform OIL REGENERATION	
	13	29/09/2017			Perform OIL REGENERATION	
	13	30/10/2017			Perform OIL REGENERATION	
	13	27/11/2017			Perform OIL REGENERATION	
	13	07/12/2017			Perform OIL PURIFICATION	
	13	21/05/2018			Perform OIL PURIFICATION	
	13	28/06/2018			Retest 12 months: Specific Gravity Retest 6 months: Dielectric and Water Content Add Passivator	
	13	24/07/2018			Perform OIL PURIFICATION	
	13	25/08/2018			Periodic condition monitoring (Purifying on end of July 2018)	
	13	25/09/2018			Perform OIL PURIFICATION	
11	14	04/07/17	Trafo 4 ESP	3	Retest 6 months: Acidity, IFT and LPF Add inhibitor 2,6-DBPC and Passivator Other parameters: Periodic Condition Monitoring	
	14	30/07/2017			Retest 6 months: Acidity, IFT and Liquid Power Factor Other parameters: Periodic Condition Monitoring	
	14	30/08/2017			Periodic condition monitoring	

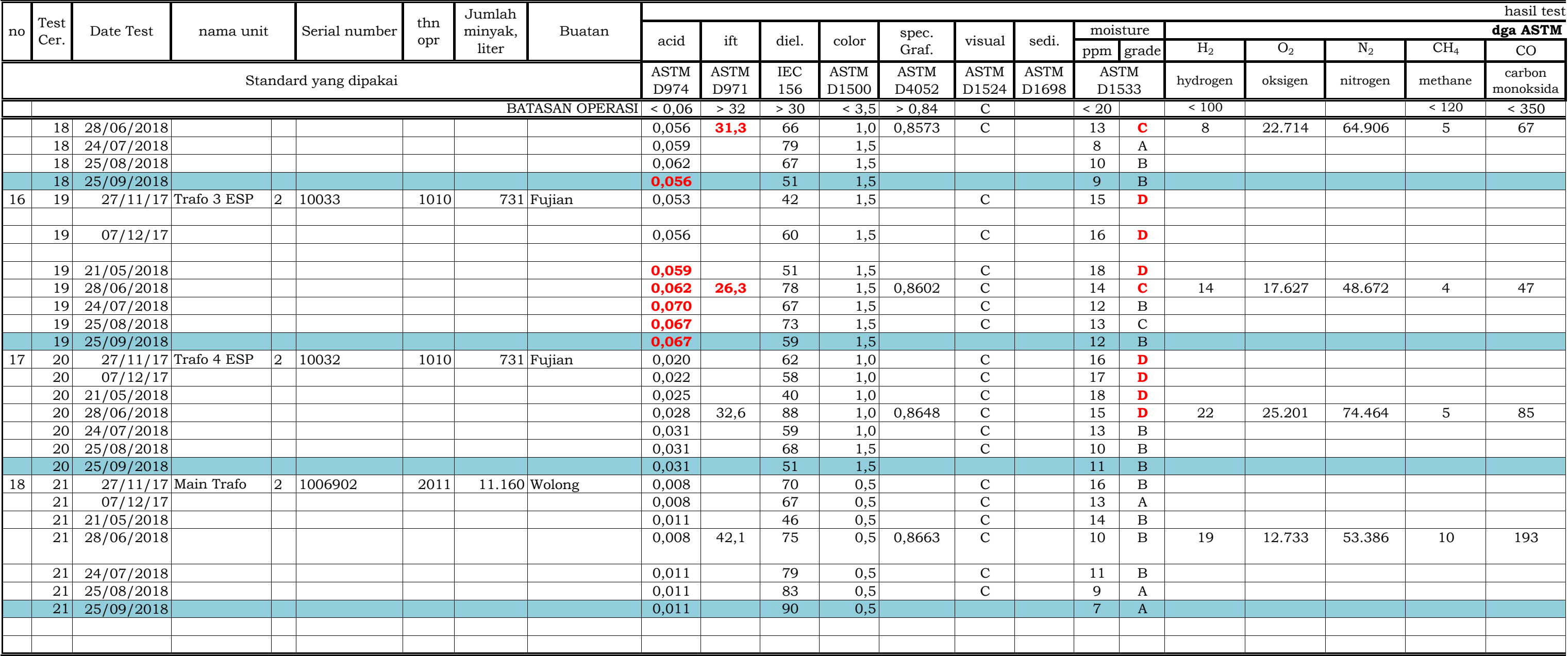


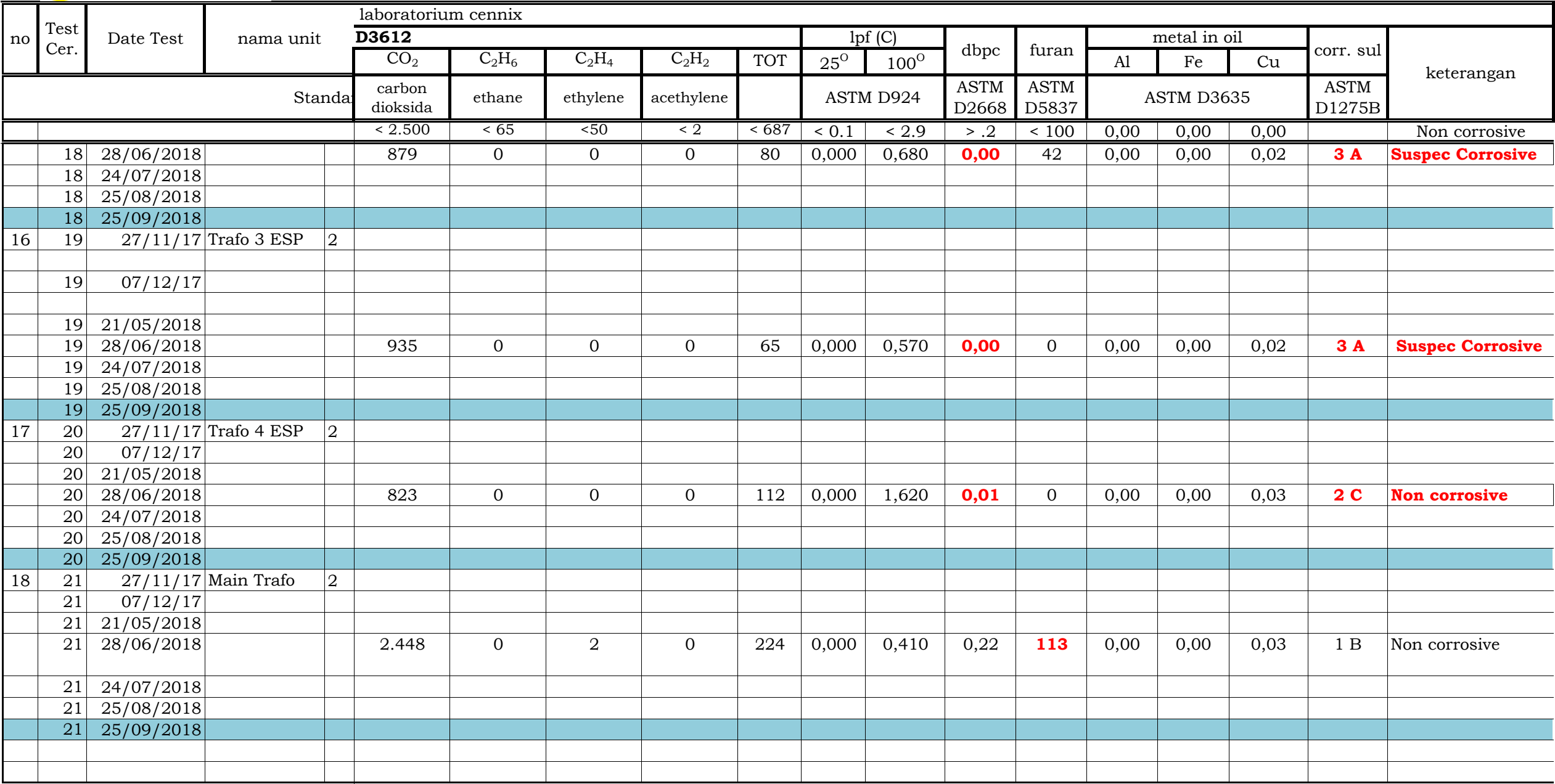
no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil test														
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture		dga ASTM					
															ppm	grade	H ₂	O ₂	N ₂	CH ₄	CO	
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533	hydrogen	oksigen	nitrogen	methane	carbon monoksida		
			BATASAN OPERASI					< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20		< 100			< 120	< 350	
	14	29/09/2017						0,022		59	1,0		C		15	C						
	14	30/10/2017						0,025		55	1,0		C		15	D						
	14	27/11/2017						0,022		69	1,0		C		16	C						
	14	07/12/2017						0,025	29,8	43	1,0	0,8471	C		19	D	14	15.414	63.712	2	22	
	14	21/05/2018						0,028		37	1,0		C		21	D						
	14	28/06/2018						0,025	30,5	71	1,0	0,8442	C		11	A	20	14.690	67.822	2	22	
	14	24/07/2018						0,022		42	1,0		C		11	C						
	14	25/08/2018						0,025		66	1,5		C		11	B						
	14	25/09/2018						0,022		51	1,5				11	B						
12	15	04/07/17	Ash Handling	10820108	2011	1.023	Wolong	0,022	38,1	38	1,0	0,8779	C		28	C	35	10.984	71.058	20	441	
	15	30/07/2017						0,025		56	1,0		C		21	D						
	15	30/08/2017						0,022		53	1,0		C		16	B						
	15	29/09/2017						0,022		57	1,0		C		16	B						
	15	30/10/2017						0,022		45	1,0		C		20	B						
	15	27/11/2017						0,020		36	1,0		C		29	C						
	15	07/12/2017						0,017	37,5	34	1,0	0,8932	C		26	B	49	12.034	67.124	20	558	
	15	21/05/2018						0,022		33	1,0		C		26	C						
	15	28/06/2018						0,025	37,5	42	1,0	0,8804	C		24	C	53	10.939	69.394	20	557	
	15	24/07/2018						0,022		39	1,0		C		20	B						
	15	25/08/2018						0,025		60	1,5		C		16	B						
	15	25/09/2018						0,025		43	1,5				17	A						
13	16	04/07/17	Ash Handling	10820107	2011	1.023	Wolong	0,022	37,3	36	1,0	0,8788	C		21	C	18	21.157	64.224	4	129	
	16	30/07/2017						0,022		40	1,0		C		32	D						
	16	30/08/2017						0,022		31	1,0		C		29	C						
	16	29/09/2017						0,022		28	1,5		C		31	B						
	16	30/10/2017						0,020		46	1,5		C		21	C						
	16	27/11/2017						0,022		47	1,5		C		20	C						
	16	07/12/2017						0,020	36,6	24	1,5	0,8940	C		32	D	20	15.791	63.530	5	275	
	16	21/05/2018						0,020		24	1,5		C		32	D						
	16	28/06/2018						0,022	35,0	50	1,5	0,8811	C		16	B	32	22.904	67.139	4	249	
	16	24/07/2018						0,020		27	1,5		C		30	C						
	16	25/08/2018						0,022		45	1,5		C		16	B						
	16	25/09/2018						0,022		36	1,5				19	B						
14	17	27/11/17	Trafo 1 ESP	2	10037	1010	731	Fujian	0,020		48	1,0		C		21	D					
	17	07/12/17						0,022		22	1,0		C		35	E						
	17	21/05/2018						0,025		24	1,0		C		32	E						
	17	28/06/2018						0,025	32,9	77	1,0	0,8693	C		11	B	93	17.225	43.231	5	57	
	17	24/07/2018						0,028		69	1,0		C		11	B						
	17	25/08/2018						0,031		74	1,5		C		12	B						
	17	25/09/2018						0,028		74	1,5				10	B						
15	18	27/11/17	Trafo 2 ESP	2	10036	1010	731	Fujian	0,048		55	1,0		C		13	C					
	18	07/12/17						0,051		59	1,0		C		14	C						
	18	21/05/2018						0,053		55	1,0		C		16	C						





no	Test Cer.	Date Test	nama unit	MALAYSIAN LAB RECOMMENDATION		PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION
Standards						
	14	29/09/2017			Periodic condition monitoring	
	14	30/10/2017			Periodic condition monitoring	
	14	27/11/2017			Periodic condition monitoring	
	14	07/12/2017			Retest 6 months: Acidity, IFT	
	14	21/05/2018			Retest 6 months: Dielectric and Water Content	
	14	28/06/2018			Add inhibitor 2,6-DBPC and PASSIVATOR	
	14	24/07/2018			Periodic condition monitoring	
	14	25/08/2018			Periodic condition monitoring (Purifying on end of July 2018)	
	14	25/09/2018			Periodic condition monitoring	
12	15	04/07/17	Ash Handling		Add Passivator	
					Periodic Condition Monitoring	
	15	30/07/2017			Periodic condition monitoring	
	15	30/08/2017			Periodic condition monitoring	
	15	29/09/2017			Periodic condition monitoring	
	15	30/10/2017			Periodic condition monitoring	
	15	27/11/2017			Periodic condition monitoring	
	15	07/12/2017			Periodic condition monitoring	
	15	21/05/2018			Periodic condition monitoring	
	15	28/06/2018			Add passivator	
	15	24/07/2018			Periodic condition monitoring	
	15	25/08/2018			Periodic condition monitoring	
	15	25/09/2018			Periodic condition monitoring	
13	16	04/07/17	Ash Handling		Add Passivator	
					Periodic Condition Monitoring	
	16	30/07/2017			Retest 6 months: Dielectric and Water Content	
					Other parameters: Periodic Condition Monitoring	
	16	30/08/2017			Periodic Condition Monitoring	
	16	29/09/2017			Retest 6 months: Dielectric and Water Content	
	16	30/10/2017			Retest 6 months: Dielectric and Water Content	
	16	27/11/2017			Periodic Condition Monitoring	
	16	07/12/2017			Perform OIL PURIFICATION	
	16	21/05/2018			Perform OIL PURIFICATION	
	16	28/06/2018			Retest 6 months: DGA	
					Add Passivator	
	16	24/07/2018			Perform OIL PURIFICATION	
	16	25/08/2018			Periodic condition monitoring (Purifying on end of July 2018)	
	16	25/09/2018			Periodic condition monitoring	
14	17	27/11/17	Trafo 1 ESP	2	Retest 6 months: Dielectric and Water Content	
					Other parameters: Periodic Condition Monitoring	
	17	07/12/17			OIL PURIFICATION	
	17	21/05/2018			OIL PURIFICATION	
	17	28/06/2018			Add inhibitor 2,6-DBPC and PASSIVATOR	
	17	24/07/2018			Periodic condition monitoring	
	17	25/08/2018			Periodic condition monitoring	
	17	25/09/2018			Periodic condition monitoring	
15	18	27/11/17	Trafo 2 ESP	2	Periodic condition monitoring	
	18	07/12/17			Retest 6 months: Acidity and IFT	
					Periodic condition monitoring	
	18	21/05/2018			Retest 6 months: Acidity and IFT	







no	Test Cer.	Date Test	nama unit	MALAYSIAN LAB RECOMMENDATION		PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION
Standar						
	18	28/06/2018			Add inhibitor 2,6-DBPC and PASSIVATOR	
	18	24/07/2018			Periodic condition monitoring	
	18	25/08/2018			Periodic condition monitoring	
	18	25/09/2018			Perform OIL REGENERATION	
16	19	27/11/17	Trafo 3 ESP	2	Retest 6 months: Acidity and IFT	
					Other parameters: Periodic Condition Monitoring	
	19	07/12/17			Retest 6 months: Acidity and IFT	
					Other parameters: Periodic Condition Monitoring	
	19	21/05/2018			Retest 6 months: Acidity and IFT	
	19	28/06/2018			Perform OIL REGENERATION	
	19	24/07/2018			Perform OIL REGENERATION	
	19	25/08/2018			Perform OIL REGENERATION	
	19	25/09/2018			Perform OIL REGENERATION	
17	20	27/11/17	Trafo 4 ESP	2	Periodic condition monitoring	
	20	07/12/17			Periodic condition monitoring	
	20	21/05/2018			Periodic condition monitoring	
	20	28/06/2018			Add inhibitor 2,6-DBPC and PASSIVATOR	
	20	24/07/2018			Periodic condition monitoring	
	20	25/08/2018			Periodic condition monitoring	
	20	25/09/2018			Periodic condition monitoring	
18	21	27/11/17	Main Trafo	2	Periodic condition monitoring	
	21	07/12/17			Periodic condition monitoring	
	21	21/05/2018			Periodic condition monitoring	
	21	28/06/2018			Retest 6 months: Furan and DGA Add Passivator	
	21	24/07/2018			Periodic Condition monitoring	
	21	25/08/2018			Periodic condition monitoring	
	21	25/09/2018			Periodic condition monitoring	

Summary by Soetjahjono
TxM Indonesia